

Mapping kinase domain resistance mechanisms for the MET receptor tyrosine kinase via deep mutational scanning

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This manuscript provides an **important** overview of potential resistance mutations within MET Receptor Tyrosine Kinase. The evidence supporting the findings is **convincing** - it should be pointed out that the approach is comparatively new for the application of protein kinases and the results are therefore of potentially great value. The results will be of value for clinicians facing drug resistance mutations, computational biologists who are training models of drug resistance mechanisms and biologists with an interest in cell signaling.

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Abstract

Mutations in the kinase and juxtamembrane domains of the MET Receptor Tyrosine Kinase are responsible for oncogenesis in various cancers and can drive resistance to MET-directed treatments. Determining the most effective inhibitor for each mutational profile is a major challenge for MET-driven cancer treatment in precision medicine. Here, we used a deep mutational scan (DMS) of ~5,764 MET kinase domain variants to profile the growth of each mutation against a panel of 11 inhibitors that are reported to target the MET kinase domain. We validate previously identified resistance mutations, pinpoint common resistance sites across type I, type II, and type I ½ inhibitors, unveil unique resistance and sensitizing mutations for each inhibitor, and verify non-cross-resistant sensitivities for type I and type II

inhibitor pairs. We augment a protein language model with biophysical and chemical features to improve the predictive performance for inhibitor-treated datasets. Together, our study demonstrates a pooled experimental pipeline for identifying resistance mutations, provides a reference dictionary for mutations that are sensitized to specific therapies, and offers insights for future drug development.

Introduction

Receptor Tyrosine Kinases (RTKs) are critical signaling molecules that activate and regulate cellular pathways. Disruption of typical RTK regulatory mechanisms through point mutations, gene amplification, protein fusions, or autocrine loops can drive the development, maintenance, and spread of cancers. Small molecule inhibitors are designed to disrupt aberrant signaling cascades by selectively targeting the kinase domain, with tyrosine kinase inhibitors (TKIs) like imatinib showing durable treatment outcomes (Cohen et al., 2021 [DOI](#); Attwood et al., 2021 [DOI](#)). Inhibitors are generally designed against either a wild-type kinase or a specific mutational profile, yet acquired mutations can alter sensitivity to different inhibitors and undermine efficacy. The most extreme case of this is resistance, which emerges in the treatment of many cancers by TKI selective pressure (Cohen et al., 2021 [DOI](#); Attwood et al., 2021 [DOI](#)). These mutations may act by altering kinase stability, expression, conformation, or activity of the target kinase. Although several recurrent mutations at inhibitor-interacting positions have predictable resistance, specific and rare resistance mutations can be associated with the interactions or conformations unique to certain inhibitors.

An attractive strategy to counter resistance is optimizing the interactions that differ between inhibitors. Small-molecule kinase inhibitors fall into four distinct groups, characterized by their binding modality to the ATP pocket and conformational preferences (Arter et al., 2022 [DOI](#); Attwood et al., 2021 [DOI](#); Zuccotto et al., 2009). Among these groups, three are ATP-competitive: type I, type II, and type I $\frac{1}{2}$ (Figure 1A-C [DOI](#)). Type I inhibitors occupy the adenosine binding pocket, form hydrogen bonds with “hinge” region residues, and favor an active conformation. Type II inhibitors also occupy the adenosine pocket but extend into an opening in the R-spine that is accessible in an inactive conformation (Arter et al., 2022 [DOI](#)) (Figure 1B [DOI](#)). Type I $\frac{1}{2}$ inhibitors combine features from both type I and type II inhibitors, engaging with both the adenosine pocket and the R-spine pocket (Arter et al., 2022 [DOI](#)) (Figure 1B [DOI](#)). Finally, type III inhibitors are allosteric, non-ATP competitive inhibitors (Arter et al., 2022 [DOI](#)) (Figure 1B [DOI](#)). Given the chemical interaction differences and conformational preferences among TKI groups for distinct kinase states, a general approach to combating resistance is sequential treatment of type I and II inhibitors (Recondo et al., 2018 [DOI](#)). However, without understanding the potential sensitivity of an acquired resistance mutation to the subsequent inhibitor, the efficacy of such strategies is not guaranteed.

The problem of which inhibitor to use and in what order is exemplified by the choice of inhibitors targeting MET kinase (Recondo et al., 2020 [DOI](#); Fernandes et al., 2021 [DOI](#)). MET is an RTK and proto-oncogene that has been implicated in the pathogenesis of gastric, renal, colorectal, and lung cancers (Frampton et al., 2015 [DOI](#); Duplaquet et al., 2018 [DOI](#); Wood et al., 2021 [DOI](#); Lu et al., 2017 [DOI](#)). Molecular profiling and next-generation sequencing of tumor samples has provided insight on cancer-associated MET variants (Frampton et al., 2015 [DOI](#); Bahcall et al., 2022 [DOI](#)). Clinical reports following post-treatment outcomes have documented resistance mutations at positions such as D1228, Y1230, G1163, L1195 for MET (Fernandes et al., 2021 [DOI](#); Lu et al., 2017 [DOI](#); Recondo et al., 2020 [DOI](#)). The challenge of acquired resistance following MET inhibitor therapy has been approached with strategies including sequential treatment of type I and type II TKIs (Recondo et al., 2020 [DOI](#); Bahcall et al., 2016 [DOI](#)), and combination therapy with type I and type II TKIs (Bahcall et al., 2022 [DOI](#); Fernandes et al., 2021 [DOI](#); Smyth et al., 2022). However, without extensive documentation of the behavior of resistance and sensitizing mutations in MET towards specific TKIs, there remains a barrier towards leveraging inhibitors for specific mutational responses,

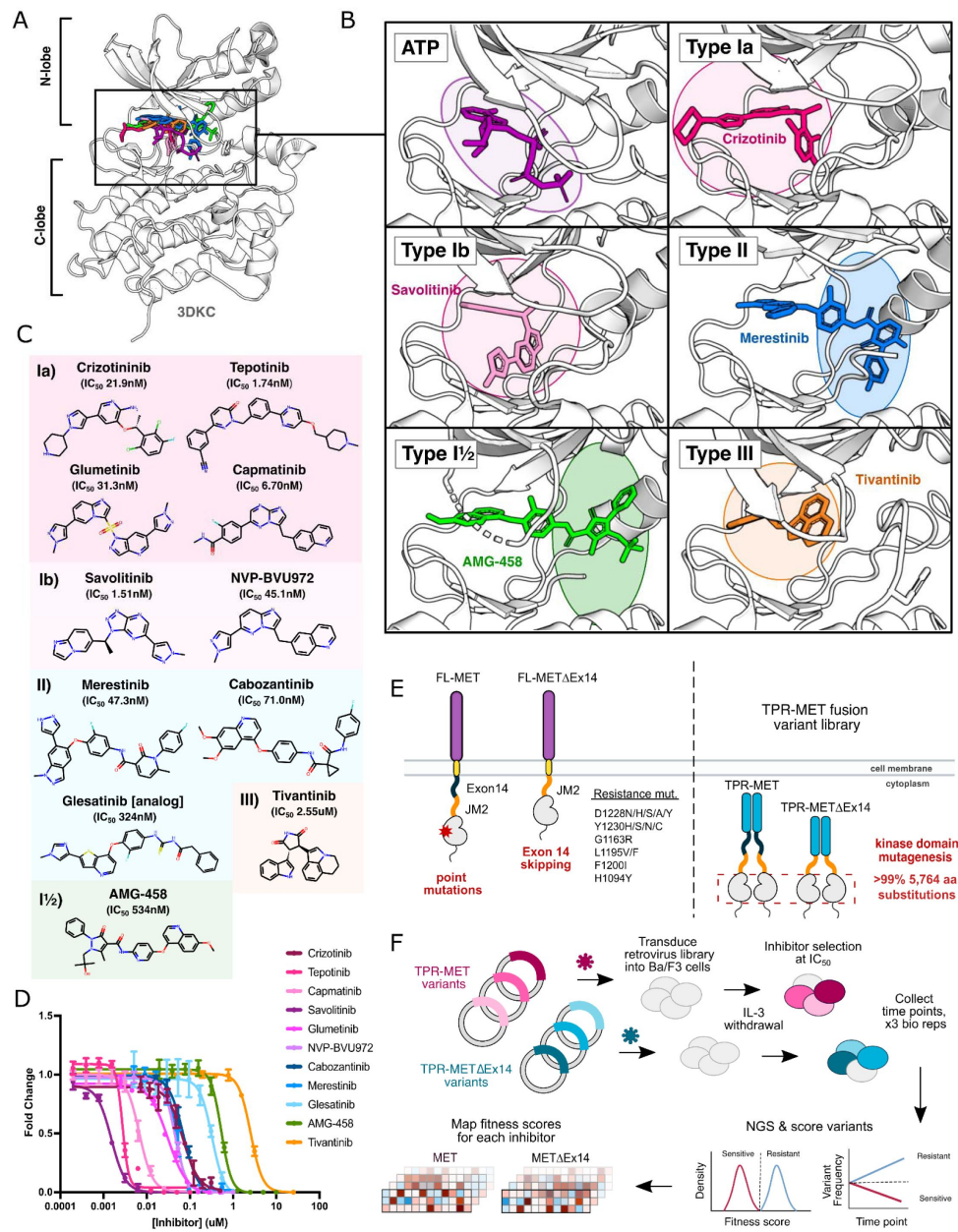


Figure 1.

MET kinase inhibitor types and resistance mutations screened against a nearly comprehensive library of kinase domain substitutions.

(A) Crystal structure of the ATP-bound MET kinase domain (3DKC) overlaid with type Ia (crizotinib, 2WGJ), type Ib (savolitinib, 6SDE), type II (merestinib, 4EEV), type I½ (AMG-458, 5T3Q), and type III inhibitors (tivantinib, 3RHK). (B) Pocket view of ATP and each inhibitor type bound to the active site of the MET kinase domain with the respective inhibitor and crystal structures from panel A. (C) 2D chemical structures of each inhibitor screened against the site saturation mutagenesis library of the MET kinase domain, with each experimentally determined IC_{50} values displayed for Ba/F3 cells stably expressing the wild-type MET ICD in a TPR-fusion background. (D) Dose-response curves for each inhibitor against the wild-type MET intracellular domain expressed in a TPR-fusion in the Ba/F3 cell line. (E) Schematics of the full-length and exon 14 skipped MET receptor alongside the TPR-fusion constructs with the full-length and exon 14 skipped intracellular domain, displaying four mechanisms of oncogenic activity: point mutations, exon 14 skipping, constitutive activity through domain fusions, and inhibitor resistance mutations. (F) Experimental workflow for defining the mutational landscape of the wild-type TPR-MET and exon 14 skipped TPR-METΔEx14 intracellular domain against 11 ATP-competitive inhibitors in Ba/F3, interleukin-3 (IL-3) withdrawn pooled competition assay.

optimizing inhibitor pairings, and informing rational drug design. Thousands of compounds have been screened against the MET kinase domain, and while several have undergone clinical trials, currently four MET inhibitors have received FDA approval: crizotinib, cabozantinib, tepotinib, and capmatinib (Santarpia et al., 2021 [↗](#); Wang and Lu, 2023 [↗](#)). Nevertheless, the emergence of resistance not only limits the efficacy of these drugs but also poses challenges for second-line therapeutic strategies, particularly in the context of rare and novel mutations.

Previously, we used Deep Mutational Scanning (DMS), a pooled cellular selection experiment, to massively screen a library of nearly all possible MET kinase domain mutations. The juxtamembrane domain is encoded by Exon 14 in MET, serves an incompletely understood negative regulatory function in the kinase (Ma et al., 2003 [↗](#)) and is recurrently excluded in cancer by somatically encoded exon skipping mutations. By testing this library in the context of a wild-type intracellular domain and the recurrent cancer exon 14 skipped variant (MET Δ Ex14) (Figure 1E [↗](#)), we identified conserved regulatory motifs, interactions involving the juxtamembrane and α C-helix, a critical β 5 motif, clinically documented cancer mutations, and classified variants of unknown significance (Estevam et al., 2024 [↗](#)). Understanding how these variants respond to specific inhibitors can inform therapeutic strategies, with precedent in inhibitor-based DMS studies across kinases such as ERK, CDK4/6, Src, EGFR, and others (Brenan et al., 2016 [↗](#); Persky et al., 2020 [↗](#); Chakraborty et al., 2023 [↗](#); An et al., 2023 [↗](#)).

Here, we explore the landscape of TKI resistance of the MET kinase domain against a panel of 11 inhibitors, utilizing our previously established platform (Estevam et al., 2024 [↗](#)). By profiling a near-comprehensive library of kinase domain variants in the MET and MET Δ Ex14 intracellular domain, we captured a diverse range of effects based on inhibitor chemistry and ‘type’ classifications (Figure 1 [↗](#)). Within our screen, mutations that confer resistance and offer differential sensitivities across inhibitors were identified, which can be leveraged in sequential or combination therapy. We use Rosace, a Bayesian fitness scoring framework, to reduce false discovery rates in mutational scoring and allow for post-processing normalization of inhibitor treatments (Rao et al., 2024 [↗](#)). With our dataset, we have analyzed differential sensitivities to inhibitor pairs and provided a platform for assessing inhibitor efficacy based on mutational sensitivity and likelihood. Lastly, we augment a protein language model (Rives et al., 2021 [↗](#); Brandes et al., 2023 [↗](#); Chen and Guestrin, 2016 [↗](#)) with biophysical and chemical features to improve predictions for MET inhibitor datasets, and in the future more effectively learn and predict mutational fitness towards novel inhibitors.

Results

Measuring the mutational fitness of 5,764 MET kinase domain variants against ATP-competitive inhibitors

To evaluate the response of MET mutations to different inhibitors, we selected six type I inhibitors (crizotinib, capmatinib, tepotinib, glumetinib, savolitinib, and NVP-BVU972), three type II inhibitors (cabozantinib, glesatinib analog, merestinib), and a proposed type III inhibitor, tivantinib (Figure 1C [↗](#)). Type I MET inhibitors leverage pi-stacking interactions with Y1230 and salt-bridge formation between D1228 and K1110, and are further classified as type Ia or Ib based on whether they interact with solvent front residue G1163 (Cui 2014 [↗](#); Fujino et al., 2019 [↗](#); Wang et al., 2023) (Figure 1C [↗](#)). Here we specifically define type Ia inhibitors as having a solvent-front interaction (Cui 2014 [↗](#)), which structurally classifies tepotinib and capmatinib as type Ia based on our analysis of experimental structures and inhibitor docked models (Figure 1A-C [↗](#); Figure 1 - figure supplement 1 [↗](#)), despite classification as type Ib in other studies (Brazel et al., 2022 [↗](#); Fujino et al., 2022 [↗](#); Recondo et al., 2020 [↗](#)).

As in our previous work, we employed the Ba/F3 cell line as our selection system due to its undetectable expression of endogenous RTKs and addiction to exogenous interleukin-3 (IL-3) (Estevam et al., 2024 [DOI](#)). These properties allow for positive selection based on ectopically expressed kinase activity and proliferation in the absence of IL-3 (Daley and Baltimore, 1988 [DOI](#); Warmuth et al., 2007 [DOI](#); Koga et al., 2022 [DOI](#)). We used a TPR-MET fusion to generate IL-3 independent constitutive activity (Estevam et al., 2024 [DOI](#)). While this system affords cytoplasmic expression, constitutive oligomerization, and HGF-independent activation, features like membrane-proximal effects are lost (Cooper et al., 1984 [DOI](#); Park et al., 1986 [DOI](#); Peschard et al., 2001 [DOI](#); Rodrigues and Park, 1993 [DOI](#); Vigna et al., 1999 [DOI](#); Mak et al., 2007 [DOI](#); Pal et al., 2017 [DOI](#); Lu et al., 2017 [DOI](#); Fujino et al., 2019 [DOI](#)). The constitutive activity of TPR-MET and the reliance of Ba/F3 cells on that activity render this selection system quite sensitive for determining reduction in growth from small molecule inhibition.

We generated dose response curves for each inhibitor against wild-type TPR-MET (wild-type intracellular domain, including exon 14) and TPR-METΔEx14 (exon 14 skipped intracellular domain) constructs, stably expressed in Ba/F3 cells to determine IC₅₀ values for our system (**Figure 1D** [DOI](#); **Figure 1 - figure supplement 1** [DOI](#)). We used our previously published library harboring >99% of all possible 5,764 kinase domain (1059-1345aa) mutations in a TPR-fusion background carrying either a wild-type MET or exon 14 skipped intracellular domain (Estevam et al., 2024 [DOI](#)) (**Figure 1E** [DOI](#)). Time points were selected every two cell doublings over the course of four time points, and cells were split and maintained in the absence of IL-3 and presence of drug at IC₅₀ for each inhibitor, including a DMSO control. All samples, across all time points and replicates, were prepared for next-generation sequencing (NGS) in parallel, and sequenced on the same Illumina NovaSeq 6000 flow cell to identify variant frequencies (**Figure 1F** [DOI](#)). We then calculated variant fitness scores using Rosace (Rao et al., 2024 [DOI](#)) (**Figure 1F** [DOI](#); **Figure 1 - figure supplement 1** [DOI](#)-**4** [DOI](#)). We performed parallel analysis of the TPR-MET and TPR-METΔEx14 screens; however, we focus our analysis below on TPR-MET with the parallel and largely consistent analyses of TPR-METΔEx14 available in the supplement.

Defining the mutational landscape of resistant and sensitizing mutations for the MET kinase domain

While growth rates were experimentally controlled through equipotent dosing during selection, there was no direct way to validate this post-processing. To generate meaningful comparisons between inhibitor scores and conditions, in addition to performing downstream score subtractions, we normalized cell growth rates for each inhibitor to the growth rate observed for the DMSO population. As expected, the DMSO control population displayed a bimodal distribution with mutations exhibiting wild-type fitness centered around 0, with a wider distribution of mutations that exhibited loss- or gain-of-function effects, as defined by fitness scores with statistically significant lower or greater scores than wild-type, respectively (**Figure 2A** [DOI](#)). Also as expected, inhibitor-treated populations displayed distributions with a loss-of-function peak, representative of mutations that are sensitive to the inhibitor. Unlike DMSO, inhibitor populations were right-skewed, showing greater enrichment of gain-of-function scores at the positive tail of distributions (**Figure 2A** [DOI](#)). These population differences were exemplified by the low correlation between each inhibitor and DMSO, with capmatinib showing the greatest difference from DMSO with a Pearson's correlation of 0.45, and tivantinib standing as an outlier with a Pearson's correlation of 0.93 (**Figure 2B** [DOI](#)).

In comparing all conditions to each other, we were able to further capture differences between and within inhibitor types. Type II inhibitors displayed the greatest similarities to one another, with merestinib and the glesatinib analog having the highest correlation ($r = 0.93$) and cabozantinib and glesatinib analog showing the lowest ($r = 0.87$) (**Figure 2C** [DOI](#)). While type I inhibitors were also highly correlated, capmatinib stood out as an outlier, displaying the lowest correlations potentially due to difficulty in experimental overdosing due to its greater potency

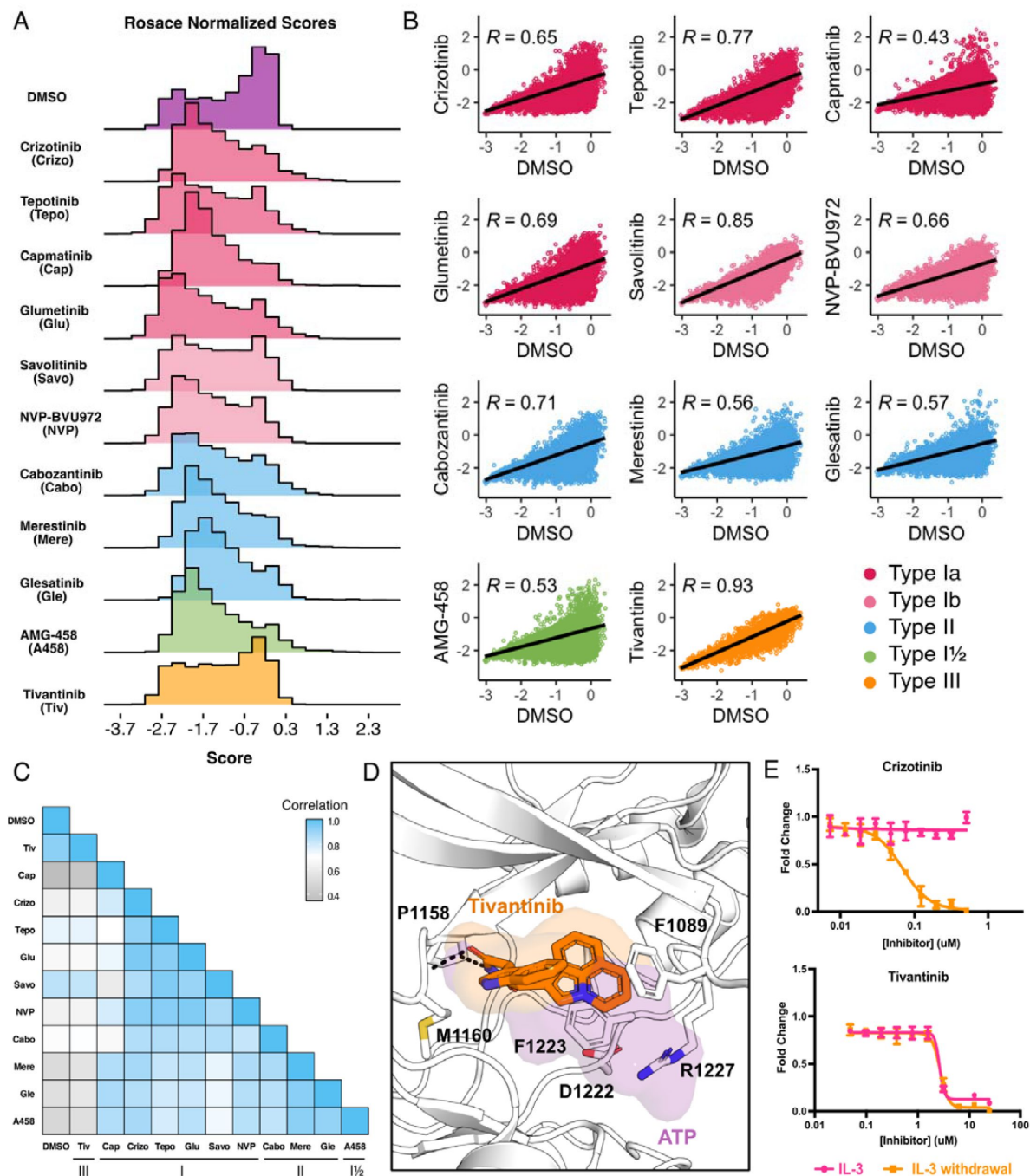


Figure 2.

Mutational landscape of the MET kinase domain under 11 ATP-competitive inhibitor selection.

(A) Distributions of all variants (wild-type synonymous, early stop, and missense) for each condition in the wild-type TPR-MET kinase domain, scored with Rosace and normalized to the growth rate of the DMSO control population. (B) Correlation plots for all mutational fitness scores for each drug against DMSO, fitted with a linear regression and Pearson's R value displayed. (C) Heatmap showing the Pearson's R correlation for each condition against each other, annotated by condition and inhibitor type. Correlations are colored according to a scale bar from gray to blue (low to high correlation). (D) Crystal structure of the tivantinib-bound MET kinase domain (PDB 3RHK) overlaid with the ATP-bound kinase domain (PDB 3DKC), with tivantinib-stabilizing residues and overlapping density of tivantinib (orange) and ATP (purple) highlighted. (E) Dose responses of crizotinib and tivantinib tested against stable Ba/F3 cells expressing the wild-type intracellular domain of MET fused to TPR, tested in the presence and absence of interleukin-3 (IL-3).

(Bahcall et al., 2022 [↗](#); Fujino et al., 2019 [↗](#)) (**Figure 2C** [↗](#)). While there was only one type I½ inhibitor, AMG-458, it displayed higher similarity to type II inhibitors than to type I, likely due to similar type II back pocket interactions with the kinase R-spine (**Figure 1B** [↗](#); **Figure 2C** [↗](#)). Nevertheless, AMG-458 was most distinct from cabozantinib ($r = 0.83$) and type I inhibitors tepotinib ($r = 0.79$) and savolitinib ($r = 0.73$) (**Figure 2C** [↗](#)). Between type I and type II groups, with the exception of capmatinib, tepotinib and savolitinib showed the lowest correlation with merestinib and glesatinib analog (**Figure 2C** [↗](#)).

The strong correlation of tivantinib with the DMSO control ($r = 0.93$) and low correlation with all other inhibitors suggested a MET-independent mode of action. Until recently (Michaelides et al., 2023 [↗](#)), tivantinib was considered the only type III MET-inhibitor and showed promising early clinical trial results (Eathiraj et al., 2011 [↗](#); Wang et al., 2022). *In vitro* assays on the purified MET kinase domain have shown that tivantinib has the potential to hinder catalytic activity (Munshi et al., 2010 [↗](#)) and structural studies revealed that it selectively targets an inactive DFG-motif conformation, with tivantinib stabilizing residues (F1089, R1227) blocking ATP binding (Eathiraj et al., 2011 [↗](#)) (**Figure 2D** [↗](#)). Yet in contradiction, comparative MET-dependent and MET-independent cell-based studies on tivantinib have also shown MET agnostic anti-tumor activity, posing that tivantinib may have an alternative inhibitory mechanism than a MET-selective one (Michieli and Di Nicolantonio, 2013 [↗](#); Basilico et al., 2013 [↗](#); Katayama et al., 2013 [↗](#); Fujino et al., 2019 [↗](#)).

Therefore, to test the hypothesis that tivantinib is not MET-selective in our system, we compared the dose response of tivantinib and crizotinib in the presence and absence of IL-3 for wild-type TPR-MET, stably expressed in Ba/F3 cells (**Figure 2E** [↗](#)). As expected, crizotinib only displayed an inhibitory effect under IL-3 withdrawal, highlighting a MET-dependent mode of action. In contrast, tivantinib displayed equivalent inhibition regardless of IL-3, reinforcing that tivantinib has cytotoxicity effects unrelated to MET inhibition (**Figure 2E** [↗](#)) and underscores the sensitivity of the DMS in identifying direct protein-drug effects.

Crizotinib-MET kinase domain resistance profiles exemplify the information accessible from individual inhibitor DMS

As an example of the insights that can be learned from the inhibitor DMS screens, we examined the profile for crizotinib, one of four FDA approved inhibitors for MET and a multitarget TKI (Cui et al., 2011 [↗](#); Wang and Lu, 2023 [↗](#); Santarpia et al., 2021 [↗](#)). To identify mutations that show gain-of-function and loss-of-function behaviors specific to inhibitors compared to DMSO, we subtracted DMSO from all fitness scores. (**Figure 3A** [↗](#)), with the expectation that effects related to expression or stability would be similar in both conditions, enhancing the ability to identify drug sensitivity or resistance. Indeed, the highest frequency of gain-of-function mutations occurred at residues mediating direct drug-protein interactions, such as D1228, Y1230, and G1163. These sites, and many of the individual mutations, have been noted in prior reports, such as: D1228N/H/V/Y, Y1230C/H/N/S, G1163R (Fernandes et al., 2021 [↗](#); Yao et al., 2023 [↗](#); Bahcall et al., 2022 [↗](#); Recondo et al., 2020 [↗](#); Rotow et al., 2020 [↗](#); Fujino et al., 2019 [↗](#); Lu et al., 2017 [↗](#); Pecci et al. 2024 [↗](#)). Beyond these well characterized sites, regions with sensitivity occurred throughout the kinase, primarily in loop-regions which have the greatest mutational tolerance in DMSO, but do not provide a growth advantage in the presence of an inhibitor.

In mapping positions with resistance to the crizotinib-bound kinase domain crystal structure (PDB 2WGJ), our DMS results further emphasize the emergence of resistance mutations at the ATP-binding site and direct-protein drug interacting residues (**Figure 3B-D** [↗](#)). Mutations to the hinge position, Y1159, and C-spine residues, including M1211 and V1092, introduce charge or are predicted to change the conformation of the pocket to clash with crizotinib but not ATP (**Figure 3E,F** [↗](#)). Outside of direct drug-protein interactions, positions I1084, T1261, Y1093, and G1242 displayed the largest resistance signals (**Figure 3A-D** [↗](#)). Structurally, I1084 is located in $\beta 1$ at the

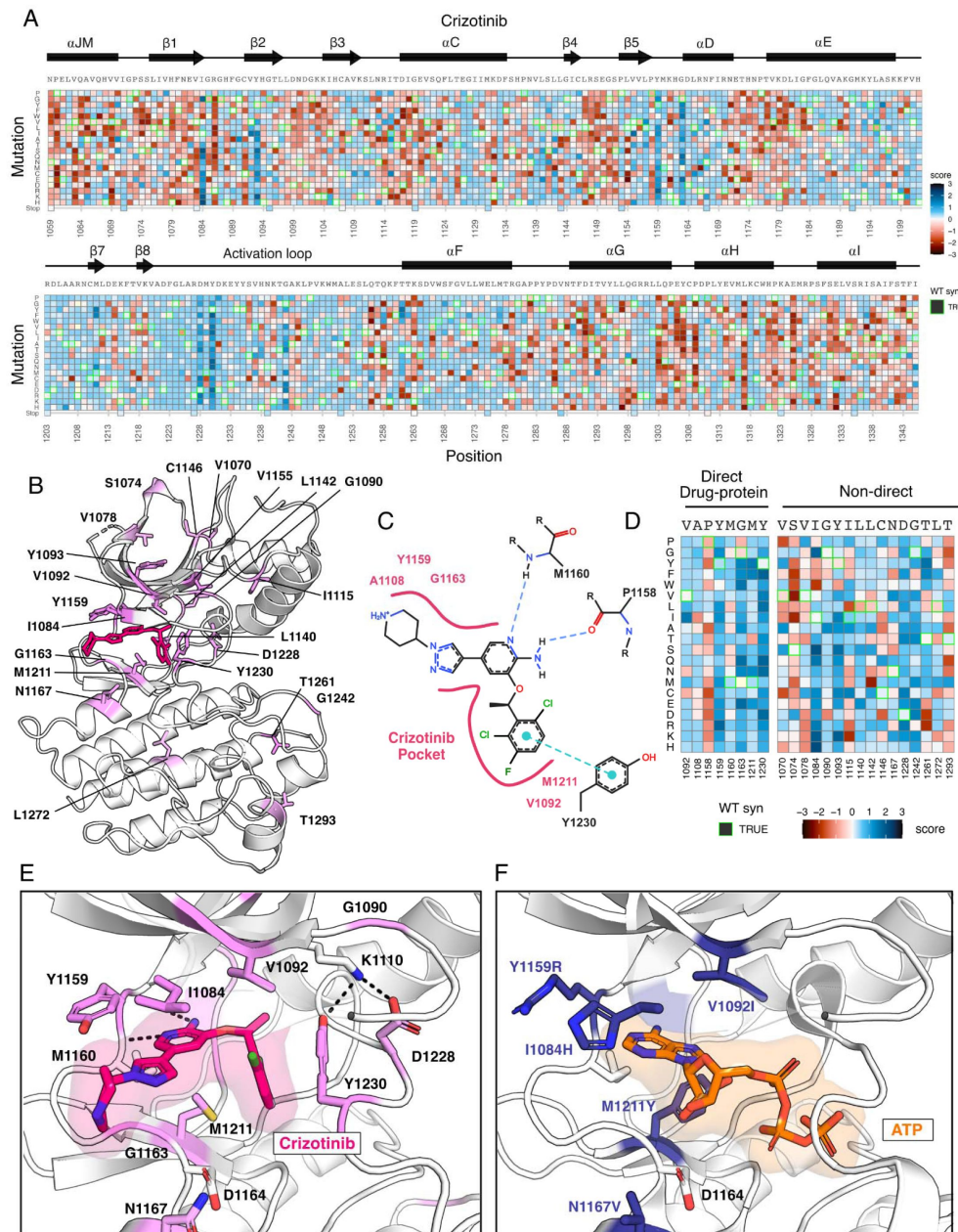


Figure 3.

Novel resistance mutations identified and mapped for crizotinib.

(A) Heatmap of crizotinib fitness scores subtracted from DMSO, scaled from loss-of-function (red) to gain-of-function (blue), with the wild-type protein sequence, secondary structure, kinase domain residue position, and mutational substitution annotated. Wild-type synonymous substitutions are outlined in green, and uncaptured mutations are in light yellow. (B) Resistance positions mapped onto the crizotinib-bound, MET crystal structure (PDB 2WGJ). Positions that contain one or multiple resistance mutations are labeled and colors are scaled according to the average score for the resistance mutations at each site. (C) 2D protein-drug interactions between crizotinib and the MET kinase domain (PDB 2WGJ) with pocket residues and polar and pi interactions annotated. Schematic generated through PoseEdit (Diedrich et al., 2023) (<https://proteins.plus/>). (D) Condensed crizotinib heatmap displaying direct drug-protein interacting and non-direct resistance position. Again, fitness scores are scaled from loss-of-function (red) to gain-of-function (blue), wild-type synonymous substitutions are outlined in green, and uncaptured mutations are in light yellow. (E) Crizotinib binding site and pocket residues displayed with resistance positions highlighted (pink) and the wild-type residue and inhibitor interactions shown (PDB 2WGJ). (F) Resistance mutations modeled for I1084H, V1092I, Y1159R, M1211Y, and N1167V relative to ATP (PDB 3DKC).

roof of the ATP-binding pocket, and a mutation to His clashes with crizotinib's hinge-binding and solvent-front moieties without interfering with bound-ATP (**Figure 3E,F**). Y1093 is also at the roof of the ATP-binding site, residing in $\beta 2$ (**Figure 3B**). However, its structural influence on resistance is unobvious. In all rotameric states, the R-group of Y1093 points away from the catalytic site and does not clash with crizotinib. We speculate that Y1093 mutations may negatively impact crizotinib's stability in the catalytic site compared to ATP, as ATP's triphosphate group is stabilized by the P-loop. Therefore, comparing crizotinib to DMSO highlights both known hotspots and rarer sites like I1084 and Y1093, which may contribute to resistance through conformational changes rather than disrupting direct inhibitor-protein contacts. Individual inhibitor resistance landscapes also aid in identifying target residues for novel drug design by providing insights into mutability and known resistance cases. This enables the selection of vectors for chemical elaboration with a potential lower risk of resistance development. Sites with mutational profiles such as R1086 and C1091, located in the common drug target P-loop of MET, could be likely candidates for crizotinib.

Resistance mutations identified for type I, type II, and type I $\frac{1}{2}$ inhibitors

To assess the agreement between our DMS and previously annotated resistance mutations, we compiled a list of reported resistance mutations from recent clinical and experimental studies (Pecci et al. 2024; Yao et al., 2023; Bahcall et al., 2022; Recondo et al., 2020; Rotow et al., 2020; Fujino et al., 2019) (**Figure 4A,B**). Overall, previously discovered mutations are strongly shifted to a GOF distribution for the drugs where resistance is reported from treatment or experiment; in contrast, the distribution is centered around neutral for those sites for other drugs not reported in the literature (**Figure 4C**). However, even in cases such as L1195V, we observe GOF DMS scores indicative of resistance to previously reported inhibitors. Given this overall strong concordance with prior literature and clinical results, we can also provide hypotheses to clarify the role of mutations that are observed in combination with others. For example, H1094Y is a reported driver mutation that has been linked to resistance in MET Δ Ex14 for glesatinib with either the secondary L1195V mutation or in isolation (Recondo et al., 2020). However, in our assay H1094Y demonstrated slight sensitivity to glesatinib, suggesting that either resistance is linked to the exon14 deletion isoform, the L1195V mutation, or a cellular factor not modeled well by the BaF3 system.

With this validation, we next wanted to identify the strongest potential resistance mutations. We identified unique resistance mutations enriched at the ATP-binding site across all inhibitors, yet also noticed discernible differences between type I and II inhibitors, the R-spine, and α C-helix (**Figure 5A-G**). Mapping inhibitor-specific positions and mutational scores, not only provides a mutation-level breakdown of inhibitor contributions to common resistance mutations, but also demonstrates differences in structural resistance enrichment across specific inhibitors (**Figure 5A-G**). To summarize this information, we next examined trends by grouping inhibitors by type.

Next, we filtered resistance mutations by their score and test statistics (**Figure 6 - figure supplement 1**) and collapsed the information by inhibitor type, plotting the total frequency of resistance mutations at each position (**Figure 6A**). In this condensed heatmap, several common resistance positions emerged within and across inhibitor types to provide a broad view of "hotspots." Two positions stood out with the highest frequency of resistance: G1163 and D1228 (**Figure 6A-D**). Both sites are unsurprising due to their inhibitor interactions - G1163 is at the solvent front entrance of the active site and D1228 stabilizes an inactive conformation of the A-loop with an inhibitor bound (Cui 2014; Recondo et al., 2020; Fernandes et al., 2021). Located at the base of the active site, M1211 is a previously documented resistance site (Tidet et al., 2011) and a C-spine residue (Estevam et al., 2024), which harbors a smaller number of resistance mutations for all inhibitor types within our analysis (**Figure 6A**). In contrast to these universal sites, Y1230 was a hotspot for type I and I $\frac{1}{2}$ inhibitors, but not a major resistance site for



Figure 4.

Comparison of previously reported resistance mutations with DMS fitness scores.

(A) Data table summarizing reported resistance mutations from clinical and experimental studies. Inhibitors linked to reported resistance cases are listed, along with corresponding DMSO subtracted fitness scores from the DMS. The scores are represented with a color gradient ranging from loss-of-function (red) to gain-of-function (blue), with the reported inhibitor scores underlined. (B) Residue locations of previously reported resistance mutations mapped on a representative crystal structure as blue spheres (2WGJ). (C) Histograms of fitness scores from the DMS for previously annotated resistance mutations, comparing their reported inhibitor (blue) to non-reported inhibitor scores (gray).

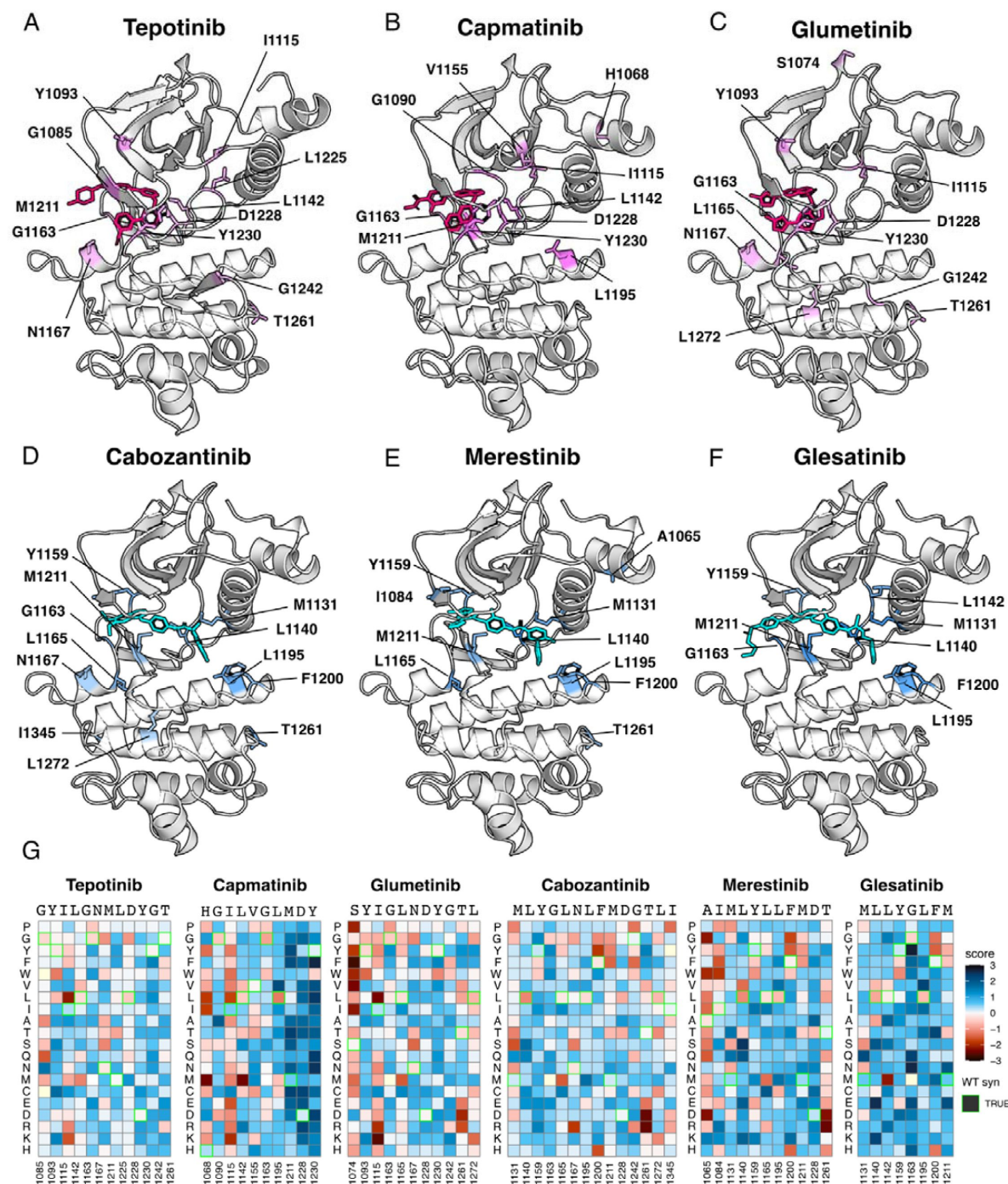


Figure 5.

Resistance mutations mapped onto experimental and docked kinase domain structures for type I and type II inhibitors.

(A-F) Resistance positions and average resistance mutational score mapped onto representative crystal structures (tepotinib, 4R1V; merestinib, 4EEV) and labeled (type I, pink; type II, blue). Inhibitors lacking experimental structures (capmatinib, cabozantinib, glumetinib, and glesatinib) were docked onto a representative type I (2WGJ) and type II (4EEV) crystal structure through AutoDock Vina (Eberheart et al., 2021; Trott et al., 2010). (G) Heatmaps of each resistance position within an inhibitor DMS. Fitness scores are scaled from loss-of-function (red) to gain-of-function (blue). Wild-type synonymous substitutions are outlined in green, and mutations uncaptured by the screen are in light yellow.

type II inhibitors (**Figure 6A-D**). This specificity can be rationalized based on the role of Y1230 in stabilizing inhibitors through pi-stacking interactions (Cui 2014). In contrast, F1200 and L1195 (Bahcall et al., 2022; Recondo et al., 2020), are both hotspots for type II but not type I inhibitors (**Figure 6A-C**). Again, this effect can be rationalized structurally: both residues make direct contact with type II inhibitors, but not type I inhibitors.

Across all inhibitor types, there were a total of 17 shared variants with G1163, D1228, and M1211 being the most common (**Figure 6G**). The overall spatial pattern of mutations for each inhibitor type follows general principles that are expected based on their interactions. For example, I1084 is enriched as a resistance site for type I inhibitors, consistent with previous studies in hereditary papillary renal cell carcinomas (Guérin et al., 2023). I1084 is located at the solvent front of the phosphate-loop (P-loop) of the kinase N-lobe (**Figure 1A**), which is responsible for stabilization of the ATP phosphate groups. This region of the kinase is leveraged for interactions with type I, but not type II inhibitors. In contrast, L1142, an R-spine residue, and L1140, which sits at the back of the ATP-binding pocket, are enriched for type II inhibitor resistance, consistent with their spatial locations (**Figure 6E-F**). Resistance mutations tend to cluster around the catalytic site across all types, but the shared mutations across different inhibitor types did not display a unifying pattern that evokes a simple rule for combining or sequencing inhibitors to counter resistance (**Figure 6H**).

Differential sensitivities of the MET kinase domain to type I and type II inhibitors

Strategies aimed at preventing resistance, such as sequential or combination dosing of type I and type II inhibitors, have been explored and offer promise in preventing resistance (Recondo et al., 2020; Bahcall et al., 2022; Fernandes et al., 2021). However, the efficacy of these strategies is limited to the emergence of secondary resistance mutations, and specific inhibitor pairings are further limited to case examples of disparate effects. Using our DMS datasets, we sought to identify inhibitor pairings with the largest divergence in cross-sensitivity. By comparing the fitness landscape of each type I inhibitor to each type II inhibitor, we could again assess inhibitor response likeness based on correlations (**Figure 2A**; **Figure 7 - figure supplement 1**). Type I and type II pairs with the highest correlations included capmatinib and glesatinib analog ($r = 0.92$), suggesting a large overlapping fitness profile, in contrast to pairs with the lower correlations, like savolitinib and merestinib ($r = 0.7$) (**Figure 7A**; **Figure 7 - figure supplement 1**). Overall, cabozantinib maintained the lowest average correlation with all type I inhibitors, making it the most divergent type II inhibitor within our screen, and potentially offering the least overlap in resistance (**Figure 7A**; **Figure 7 - figure supplement 1**).

To narrow our characterization of cross-sensitivity, we focused on the inhibitor pair crizotinib and cabozantinib (**Figure 7B**; **Figure 7 - figure supplement 1**). By statistically filtering mutations that are categorized as gain-of-function in one inhibitor, but loss-of-function in the other, a set of 44 mutations were identified as having crizotinib resistance and cabozantinib sensitivity, and 3 mutations with the opposite profile (**Figure 7B,C**). Structural mapping of divergent mutations further revealed enrichment at the N-lobe and typical protein-drug interaction sites like Y1230, G1163, and M1211 (**Figure 7B, C**). While these positions have precedence for resistance, as previously noted, they are also resistance hotspots across all inhibitor types (**Figure 6A**), where even mutations with differential sensitivities may be insufficient targets to counteract the reemergence of resistance, thus limiting the interchangeability of drugs.

Understanding which mutations have resistance profiles for only type I or type II inhibitors provides better leverage for sequential and combination dosing. To identify such mutations across our dataset, we further filtered variants that met our resistance metrics and were only observed for inhibitors of the same type. In again comparing crizotinib to cabozantinib, Y1093K was a mutation with one of the largest differences between crizotinib and cabozantinib, having a gain-

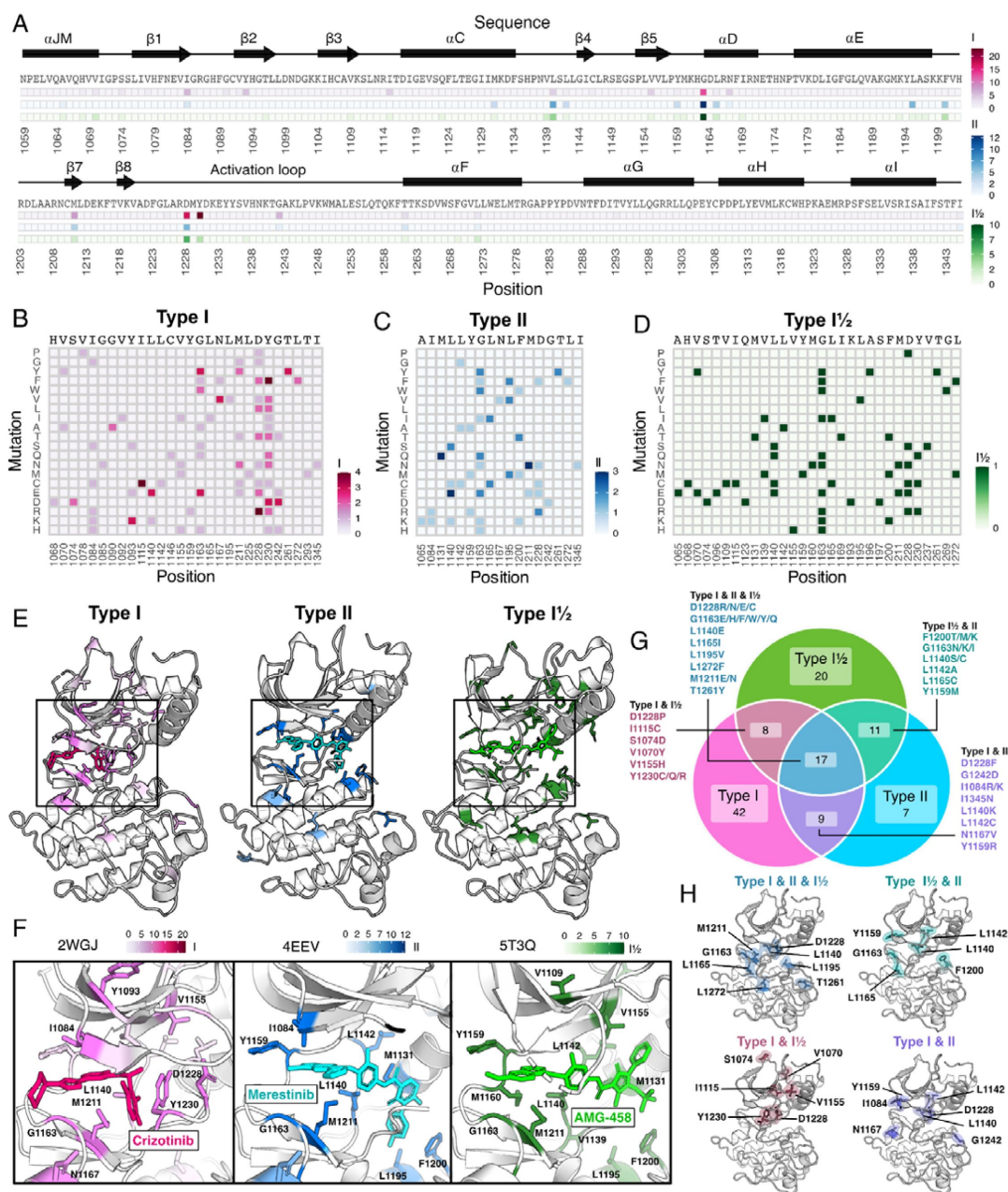


Figure 6.

Resistance mutations and “hotspots” identified for MET inhibitor types.

(A) Collapsed heatmap of common resistance positions along the kinase domain, with the wild-type protein sequence and secondary structure annotated. Each tile represents a sum of counts for statistically filtered resistance mutations across all inhibitors for type I (pink), type II (blue), and the type I½ inhibitor AMG-458 (green), with the scale bar reflecting counts of resistance mutations across respective inhibitor types. (B-D) Expanded heatmap showing each resistance position and the counts for each specific resistance mutation across all inhibitor types type I (pink), type II (blue), and the type I½ inhibitor AMG-458 (green). Wild-type sequence and variant change are annotated. (E-F) Average frequency of resistance mutations for each mapped on to a representative type I (crizotinib, 2WGJ) and type II (merestinib, 4EEV) crystal structure, alongside the type I½, AMG-458 structure (5T3Q), with associated scale bars. Individual positions with high resistance mutation frequencies are annotated on each structure, with a zoom-in of the bound inhibitor and surrounding resistance sites. (G) Venn diagram showing mutations shared among type I (pink), type II (light blue), and type I½ (green). (H) Structurally mapped (PDB 2WGJ) resistance positions shared among type I, II, I½ (blue-gray), type I and II (purple), type I and I½ (dusty rose), type II and I½ (teal) inhibitors.

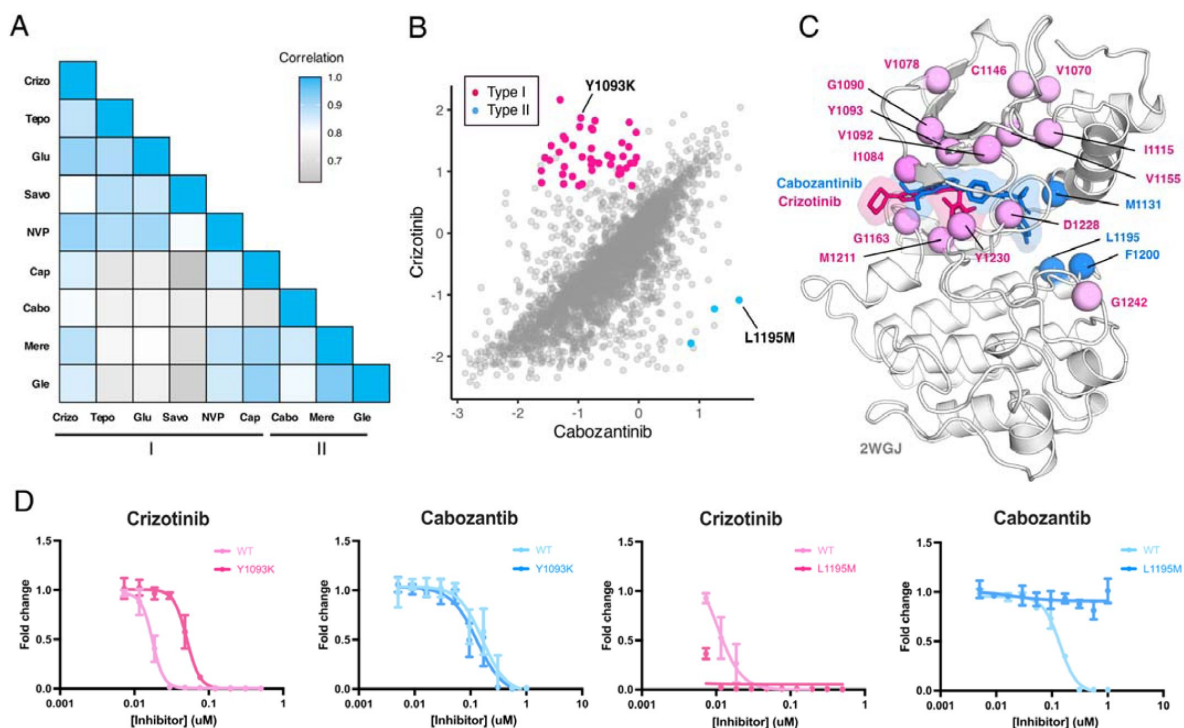


Figure 7.

MET kinase domain differential sensitivities revealed for type I and type II inhibitors.

(A) Heatmap showing Pearson correlation values for all combinations of screened type I and type II inhibitors. Correlations were determined from DMSO subtracted fitness scores (B) Correlation plot correlation plot of DMSO subtracted fitness scores for crizotinib and cabozantinib. Mutations with differential scores are highlighted for type I (pink) and type II (blue). (C) Average scores of mutations with differential sensitivities within inhibitor pairs mapped and annotated in respective crystal structures (crizotinib, 2WGJ; cabozantinib, docked into 4EEV). Positions that are gain-of-function for type I but loss-of-function in type II are highlighted in pink, whereas positions that are gain-of-function for type II but loss-of-function in type I are highlighted in blue. (D) Dose-response curves for crizotinib and cabozantinib in Ba/F3 cells expressing TPR-MET (full MET intracellular domain) harboring mutations at Y1093K and L1195M. Dose-response for each inhibitor concentration is represented as the fraction of viable cells relative to the TKI free control.

of-function profile for crizotinib and loss-of-function for cabozantinib (**Figure 7B**). Interestingly, Y1093 is located in $\beta 2$ of the N-lobe, at the roof of the ATP-binding site, and does not directly engage with crizotinib. We speculate this mutation potentially contributes to resistance by perturbing the packing of $\beta 1$ - $\beta 2$ and altering the conformation of the ATP binding site in a manner that destabilizes crizotinib binding. When comparing the dose-response of Y1093K to the wild-type TPR-MET kinase domain, Y1093K shows a nearly 10-fold shift in crizotinib sensitivity with no difference in cabozantinib sensitivity (**Figure 7D**). In identifying mutations with the opposite profiles, resistance to cabozantinib and sensitivity to crizotinib, L1195M displayed the greatest differential scores (**Figure 7B**). L1195 is an α E-helix position with previously recorded resistance (L1195V/F), which our analysis further supports as a type II-only resistance hotspot (**Figure 6A**). Structurally, mutations like Met or Phe at 1195 clash with the fluorophenyl moiety of cabozantinib used to access and stabilize a deep, back pocket of the kinase in an inactive conformation, unlike crizotinib which occupies the solvent front and adenosine binding region of the ATP binding site. In comparing the dose-response of L1195M to the wild-type TPR-MET kinase domain, we find that L1195M is refractory to all concentrations of cabozantinib tested, but still sensitive to crizotinib (**Figure 7D**). Beyond a type I and type II pairing, such cross-resistance identification can be further applied to identify differential sensitivities within an inhibitor group (**Figure 7 - figure supplement 2**), which can further expand opportunities for inhibitor-specific sensitivity in therapy and drug design.

Identification of biophysical contributors to inhibitor-specific fitness landscapes machine learning using

Machine learning models originally developed for predicting protein structure (Jumper et al., 2021; Rives et al., 2021; Lin et al., 2023) have been adapted for predicting protein-ligand complexes (Bryant et al., 2023), and predicting fitness values from DMS studies (Meier et al., 2021; Brandes et al., 2023; Jones et al., 2020). In particular, protein language models have shown the ability to estimate the functional effects of sequence variants in correlation with DMS data (Rives et al., 2021; Meier et al., 2021; Brandes et al., 2023). We observed that ESM-1b, a protein language model (Rives et al., 2021), predicts the fitness of variants in the untreated/DMSO condition (correlation 0.50) much better than it does for the inhibitor treated datasets (correlation 0.28). This difference in predictive ability is likely because the language model is trained on sequences in the evolutionary record and fitness in the presence of inhibitors does not reflect a pressure that has operated on evolutionary timescales.

To overcome this limitation and improve the predictive properties of the ESM approach, we sought to augment the model with additional features that reflect the interactions between protein and inhibitors that are not present in the evolutionary record (Chen and Guestrin, 2016). While our features can account for some changes in MET-mutant conformation and altered inhibitor binding pose, the prediction of these aspects can likely be improved with new methods. There are several challenges associated with this task, including the narrow sequence space explored, high correlations between datasets, and the limited chemical space explored by the 11 inhibitors. We used an XGBoost regressor framework and designed a test-train-validation strategy to account for these issues (**Figure 8A**), exploring many features representing conformation, stability, inhibitor-mutation distance, and inhibitor chemical information (**Figure 8 - Figure Supplement 1**). To avoid overfitting, we introduced several constraints on the monotonicity and the precision of certain features. The final model uses a subset of the features we tested and improves the performance from 0.28 to 0.37 (**Figure 8B,C**). The model primarily improves the correlation by shifting the distribution of predicted fitness values to center around drug sensitivity, reflecting the pressures that are not accounted for by ESM-1b (**Figure 8D**). Nonetheless, many resistant mutations are correctly predicted by the new model.

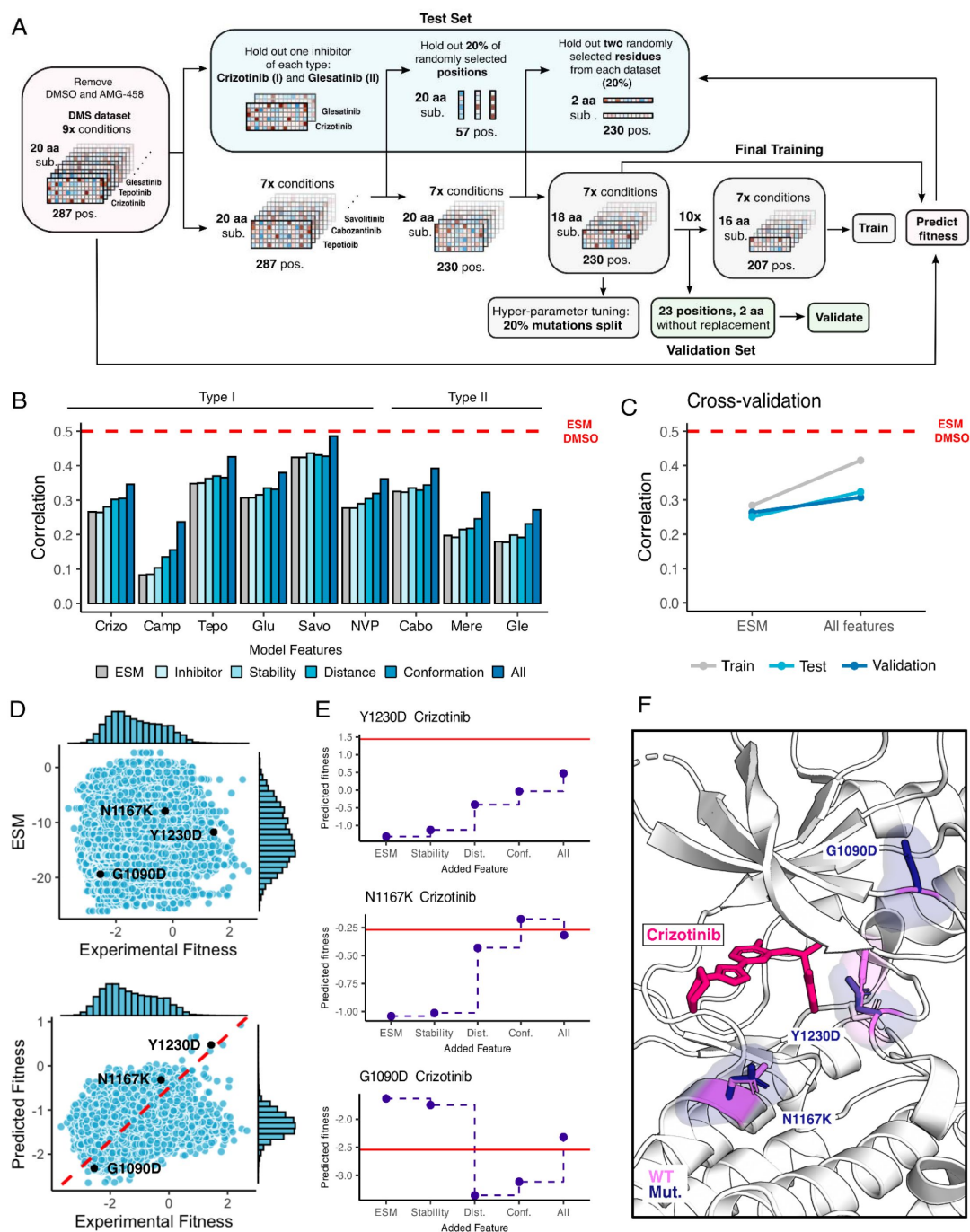


Figure 8.

Inhibitor-bound variant fitness predicted from a machine learning model trained on the MET DMS dataset.

(A) Model architecture outlining the information flow and inputs for model training, validation, fitness predictions, and prediction tests. (B) Improvement in correlation between experimental and predicted fitness for each inhibitor with usage of different kinds of features. (C) Cross-validation trends between the baseline ESM model and the model with all features incorporated. (D) Scatter plots of predictions versus experimental fitness scores of the baseline ESM model (top) compared to the model with all features (bottom), with a dashed cross-graph line in red displayed. (E) Residue-level analysis of feature significance in fitness predictions (ESM, stability, distance, conformation, all features). The Rosace experimental score is shown as a red line. (F) Residues with improved predictions mapped on a crizotinib-bound MET kinase domain (PDB 2WGJ). Predicted resistance mutations (dark purple) modeled relative to the wild-type residue (pink).

To examine whether the model could help interpret the mechanisms of specific mutations, we examined several cases with notable improved predictions as the model increased in complexity (**Figure 8E,F**). For some mutations, as in Y1230D, we observe a gradual improvement in prediction for each set of features, suggesting that resistance relies on multiple factors. For other mutations, such as N1167K, we see a single set of features driving the improvement, which suggests much more dominant driving forces. Lastly, in other mutations, like G1290D, the models trained with different features can over or under predict the true value, demonstrating the value of combining features together. The reliance on simple features helps identify some of the major factors in drug resistance and sensitization such as distance to the inhibitor and active/inactive conformation; however, improved feature engineering and coverage of both sequence and chemical space will likely be needed to create a more interpretable model.

Discussion

Tyrosine kinase inhibitors have revolutionized the treatment of many diseases, but the development of resistance creates a significant challenge for long term efficacy. Many strategies, including sequential dosing (Attwood et al., 2021; Recondo et al., 2020), are being explored to overcome resistance. Our DMS of the MET receptor tyrosine kinase domain, performed against a panel of varying inhibitors offers a framework for experimentally identifying resistance and sensitizing mutations in an activated kinase context for different inhibitors. By massively screening the effect of a nearly comprehensive library of amino acid mutations in the MET kinase domain against 11 inhibitors, some generalizable patterns emerged. In concordance with the binding mode of both type I and II inhibitors, residues that commonly confer resistance, or act as “hotspots,” were mapped to previously reported sites like D1228, Y1230, M1211, G1163 (**Figure 4**), and novel sites like I1084, L1140, L1142, T1261, and L1272 (**Figure 5**). Annotation of hotspots also offers an opportunity to inform inhibitor selection based on likelihood of cross-inhibitor resistance (**Figure 6**). For instance, I1084 is a hotspot for type I and II inhibitors within our study that displayed wild-type sensitivity to the type I½ inhibitor screened (**Figure 6**). Understanding positions with high resistance frequencies that are distal from the ATP-binding site also offers a design opportunity for allosteric inhibitors that can target cancer-associated and resistance-associated regions within the N- and C-lobe (Mingione et al., 2023).

Nevertheless, similar to its ability in identifying resistance for inhibitors, our parallel DMS also demonstrated the ability to detect non-selective drugs, with the example of tivantinib. Despite being a proposed MET-selective inhibitor, like several others, tivantinib failed clinical trials, and follow-up studies suggested cytotoxicity and off-target binding as the culprit (Michieli and Di Nicolantonio, 2013; Basilico et al., 2013; Katayama et al., 2013; Fujino et al., 2019) - a scenario that is not uncommon to antitumor drugs that do not advance to the clinic (Lin et al., 2019). To this effect, the ability of DMS to differentiate between selective compounds provides a unique prospect for developing inverse structure-activity relationships, whereby varying protein sequence both inhibitor specificity and resistance can be learned.

Reported cancer mutations in databases such as OncoKB or cBioPortal are useful for patient data and cancer type reporting (Suehnholz et al., 2024; Chakravarty et al., 2017; Cerami et al., 2012). A recent analysis of these databases aided in annotation of mutations observed within patient populations (Pecci et al., 2024). This study used pre-clinical models to examine a subset of these mutations and identified sensitivities to multiple inhibitors and confirmed clinical responses of two rare driver mutations (H1094Y and F1200I) to elzovantinib, a type Ib inhibitor (Pecci et al., 2024). Their results are consistent with the predictions of our DMS, illustrating the potential value of having a broad dictionary of inhibitor sensitivity and resistance patterns.

Finally, a significant challenge of inhibitor screening is the considerable time and cost involved, even at high-throughput. While docking has accelerated the prioritization of compounds for protein targeting and screening *in silico* (Sadybekov and Katritch, 2023 [↗](#)), prediction of drug resistance is of high interest in informing iterative drug design. Screening for resistance in the early stages of drug design is particularly useful for obtaining inhibitors that can be effective in the long-term by optimizing protein-inhibitor interactions in the wildtype and functionally silent mutant context (Pisa and Kapoor, 2020). While base-editor approaches can rapidly screen for inhibitor resistance mutations within full-length, endogenous genes, undersampling of rare variants due to lower coverage is a significant caveat (Dorigi et al., 2024 [↗](#)), compared to DMS where nearly full coverage is achieved and controlled. A full landscape of mutational effects can help to predict drug response and guide small molecule design to counteract acquired resistance. The ability to define molecular mechanisms towards that goal will likely require more purposefully chosen chemical inhibitors and combinatorial mutational libraries to be maximally informative. The ideas motivating our ML-model, which combines protein language models and biophysical/chemical features, to novel inhibitors could eventually be used to profile resistance and sensitivity for novel and unscreened small molecules, greatly extending the scale of kinase inhibitor repositioning for second-line therapies.

Materials and methods

Mammalian cell culturing

Ba/F3 cells (DSMZ) were maintained and passaged in 90% RPMI (Gibco), 10% HI-FBS (Gibco), 1% penicillin/streptomycin (Gibco), and 10ng/ml IL-3 (Fisher), and incubated at 37°C with 5% CO₂. Cells were passaged at or below 1.0e6 cells/ml to avoid acquired IL-3 resistance, and regularly checked for IL-3 dependence by performing 3x PBS (Gibco) washes and outgrowth in the absence of IL-3.

Plat-E cells stably expressing retroviral envelope and packaging plasmids were originally gifted by Dr. Wendell Lim, and maintained in 90% DMEM, HEPES (Gibco), 10% HI-FBS (Gibco), 1% penicillin/streptomycin (Gibco), 10ug/ml blasticidin, 1ug/ml puromycin. Cells were cultured at 37°C with 5% CO₂ and maintained under blasticidin and puromycin antibiotic pressure unless being transfected.

Dose response and IC₅₀ determination of inhibitors

Unless otherwise stated, all inhibitors used in this study were purchased from SelleckChem.

Ba/F3 cells stably expressing TPR-MET and TPR-METΔEx14 were washed with DPBS (Gibco) 3x times to remove IL-3, puromycin, penicillin, and streptomycin. Cells were resuspended in 90% RPMI and 10% FBS, and were seeded in the wells of a 96-well, round-bottom plate at a density of 2.5e4 cells/ml in 200ul. Cells were incubated for 24hr to allow kinase-driven signaling. The next day, inhibitors were added to triplicate rows of cells at a concentration range of 0 to 10uM (2-fold dilutions), and allowed to incubate for 72 hr post TKI addition. CellTiter-Glo reagent (Promega) was mixed at a 1:1 ratio with cells to lyse and provide a luminescence readout, which was measured on a Veritas luminometer. Cell numbers were determined from a Ba/F3 cell and ATP standard curve generated according to the manufacturer's instructions. Dose response curves were fitted using GraphPad Prism with the log(inhibitor) vs. response, variable slope function. Data are presented as cell viability normalized to the fold change from the TKI free control.

MET kinase domain variant library generation, cloning, and library introduction into Ba/F3

In this study, we repurposed cell lines transduced with TPR-MET and TPR-METΔEx14 kinase domain variant libraries, previously reported in [Estevam et al., 2024](#) [\[1\]](#). All libraries were generated, transfected, and tested in parallel.

In short, the MET kinase domain sequence used in this study spans amino acid positions 1059-1345, which includes the full kinase domain (aa 1071-1345) and a small region of the juxtamembrane (aa 1059-1070). The variant DNA library was synthesized by Twist Bioscience, containing one mammalian high-usage codon per amino acid. A “fill-in” library was generated to introduce an early stop control codon every 11 amino acids evenly spaced across the sequence. In addition, mutations at positions with failed synthesis (positions 1194 and 1278) were generated and added at equimolar concentration into the variant library. The kinase domain variant library was introduced into two different cloning backbones, one carrying the TPR-fusion sequence with the wild-type juxtamembrane sequence (aa 963-1058), wild-type C-terminal tail (aa 1346-1390), and IRES-EGFP (pUC19_kozak-TPR-METΔEx14-IRES-EGFP) and the other carrying the TPR-fusion sequence with an exon 14 skipped juxtamembrane sequence (aa 1010-1058), wild-type C-terminal tail (aa 1346-1390), and IRES-mCherry (pUC19_kozak-TPR-MET-IRES-mCherry). The libraries were transformed in MegaX 10 beta cells (Invitrogen), propagated in 50mL LB and Carbinicillin at 37°C to an OD of 0.5, and then midiprep (Zymo). Library coverage was determined by colony count of serial dilutions from the recovery plated at varying dilutions (1:100, 1:1k, 1:10k, 1:100k, 1:1M).

The full TPR-METΔEx14-IRES-EGFP and TPR-MET-IRES-mCherry variant libraries were then shuttled into the mammalian retroviral MSCV backbone (addgene) through restriction enzyme digest with MluI-HF (NEB) and MfeI-HF (NEB), then ligated into the empty backbone with T4 ligase (NEB). Ligations were DNA cleaned (Zymo), electroporated into ElectroMAX Stbl4 Competent Cells (Thermo Fisher), plated on LB-agar bioassay plates with Carbenicillin, incubated at 37°C, then colonies were scraped into 50mL LB and midi-prepped for transfections (Zymo).

Variant libraries were transfected into Plat-E cells for retroviral packaging using Lipofectamine3000 (Invitrogen) following the manufacturer’s for a T-175 scale, and using a total of 46 µg DNA. 48 hr post-transfection, the viral supernatant was harvested, passed through a 0.45 µm sterile filter, then concentrated with Retro-X concentrator (TakaraBio) using a 1:4 ratio of concentrator to supernatant. The concentrated virus was titered in Ba/F3 to determine the proper volume for a transduction MOI of 0.1-0.3. The viral titer was calculated from the percent of fluorescent cells and viral dilution. To generate the DMS transduced cell lines, 6 million cells were spininfected at an MOI of 0.1, in triplicate. Then infected cells were selected with 1µg/ml puromycin in 4 days, with fluorescence and cell counts tracked each day.

DMS time point selection and sample preparation

All screening conditions were performed and handled in parallel for TPR-MET and TPR-METΔEx14 libraries across all independent conditions and biological replicates.

For each biological replicate, a stock of 4.0e6 cells transduced with TPR-MET and TPR-METΔEx14 kinase domain variants was thawed and expanded for 48 hrs in the presence of IL-3 and puromycin to prevent pre-TKI selection to reach a density for screen seeding. Each batch of cells were grown to a density of 72 million cells to be split into 12 dishes (15cm) for each selection condition. Cells were first washed with DPBS (Gibco) 3 times to remove IL-3 and antibiotics. Cells were resuspended in 90% RPMI and 10% FBS, counted, and split across 12 dishes (15cm) at a density of 6 million cells in 30mL. A total of 6 million cells from each replicate was harvested and pelleted at 250xg to serve as the “time point 0” pre-selection sample (T0).

To begin selection of each replicate for each library, DMSO was added to the control plate (0.01% final) while the appropriate IC50 concentration of inhibitor was added to each respective plate (independent pool of cells). Three time points post T0 were collected for each library replicate and inhibitor condition for a total of 4 time points (T0, T1, T2, T3). Time points were harvested every two doublings (~72hr) across 12 days; 6 million cells were harvested for each condition and pelleted at 250xg for 5min; 2.0e5 cells/ml were split at every time point and maintained either in DMSO or TKI at the appropriate concentration to maintain cellular growth rates under inhibitor selection.

The gDNA of each time point sample was isolated with the TakaraBio NucleoSpin Blood QuickPure kit the same day the cells were harvested. gDNA was eluted in 50µl of elution buffer provided by the kit, using the high concentration and high yield elution manufacturer's protocol. Immediately after gDNA was isolated, 5µg of gDNA was used for PCR amplification of the target MET KD gene to achieve the proper variant coverage. A 150µl PCR master mix was prepared for each sample using the TakaraBio PrimeStar GXL system according to the following recipe: 30µl 5X PrimeStar GXL buffer, 4.5µl 10µM forward primer (0.3uM final), 4.5µl 10µM reverse primer (0.3uM final), 5µg gDNA, 12µl 10mM dNTPs (2.5mM each NTP), 6µl GXL polymerase, nuclease free water to a final reaction volume of 150uL. The PCR master mix for each sample was split into three PCR tubes with 50µl volumes for each condition and amplified with the following thermocycler parameters: initial denaturation at 98°C for 30 s, followed by 24x cycles of denaturation at 98°C for 10 s, annealing at 60°C for 15 s, extension at 68°C for 14 s, and a final extension at 68°C for 1 min.

PCR samples were stored at -20°C until all time points and replicates were harvested and amplified, so as to prepare all final samples for NGS together with the same handling and sequence them in the same pool to prevent sequencing bias.

Library preparation and Next-generation sequencing

After all time points were selected, harvested, and PCR amplified, the target gene amplicon was isolated from gDNA by gel purification (Zymo), for a total of 222 samples. The entire 150µl PCR reaction for each sample was mixed with 1X NEB Purple Loading Dye (6X stock) and run on a 0.8% agarose, 1X TBE gel, at 100 mA until there was clear ladder separation and distinct amplicon bands. The target amplicons were gel excised and purified with the Zymo Gel DNA Recovery kit. To remove excess agarose contamination, each sample was then further cleaned using the Zymo DNA Clean and Concentrator-5 kit and eluted in nuclease free water. Amplicon DNA concentrations were then determined by Qubit dsDNA HS assay (Invitrogen).

Libraries were then prepared for deep sequencing using the Nextera XT DNA Library Prep kit in a 96-well plate format (Illumina). Manufacturer's instructions were followed for each step: tagmentation, indexing and amplification, and clean up. Libraries were indexed using the IDT for Nextera Unique Dual Indexes Set A,B and C (Illumina). Then, indexed libraries were quantified using the Agilent TapeStation with HS D5000 screen tape (Agilent) and reagents (Agilent). DNA concentrations were further confirmed with a Qubit dsDNA HS assay (Invitrogen). All samples were manually normalized and pooled at 10nM (MET and METΔEx14 in the same pool). The library was then paired-end sequenced (SP300) on two lanes of a NovaSeq6000.

MET kinase domain variant analysis and scoring

Enrich2 Scoring

Our approach followed the one used for our initial MET DMS experiments (Estevam et al., 2024 [\[1\]](#)). Sequencing files were obtained from the sequencing core as demultiplexed fastq.gz files. The reads were first filtered for contamination and adapters using BBDuk, then the paired reads were error-corrected and merged with BBMerge and mapped to the reference sequence using BMap

(all from BBTools) (**BBMap – Bushnell B.** – sourceforge.net/projects/bbmap/). Read consequences were determined and counted using the AnalyzeSaturationMutagenesis tool in GATK v4 (van der Auwera and O'Connor, 2020). This is further processed by use of a script to filter out any variants that are not expected to be in the library (i.e., variants due to errors in sequencing, amplification, etc). The final processed count files were then analyzed with Enrich2, using weighted least squares and normalizing to wildtype sequences (NCBI SRA BioProject PRJNA1136906).

Rosace scoring

We used Rosace to analyze experiments of different conditions (DMSO or inhibitors) independently. In order to make the scores more comparable and interpretable across conditions, we modified the original Rosace software so that the output scores reflect the scale of cell doubling rate between every contiguous time point. For example, in an ideal experiment, if the wild-type cell doubling rate is 2 and its score is 0 by wild-type normalization, a score of -2 means that the cells are not growing ($2^{-(2-0)}$) and a score of 1 means that the cells are doubling three times (increasing to $2^{(2+1)}$ times the original count) between every contiguous time point.

The input to Rosace is the filtered count files provided by the AnalyzeSaturationMutagenesis tool described in the above section. From here, we filtered variants by the mean count (≥ 4) and the proportion of 0 count across replicates and time points ($\leq 10/12$) and before inhibitor selection at T0 ($\leq 2/3$). Second, we normalized the counts using wild-type normalization with log2 transformation rather than the default natural log transformation to maintain the doubling rate scale. Finally, the normalized count is regressed on time intervals ($t = \{1, 2, 3, 4\}$) instead of the entire time span ($t = \{1/4, 2/4, 3/4, 4/4\}$) so that the resulting score reflects the growth rate between every contiguous time point.

Statistical filtering and resistance classification

In mathematical terms, we define the raw Rosace fitness scores of a mutation in DMSO as $\beta_{v,DMSO}$ and in a specific inhibitor condition as $\beta_{v,inh}$. The scores of wildtype variants were normalized to 0, and we denote them as $\beta_{wt,DMSO} = 0$ and $\beta_{wt,inh} = 0$. Growth rate of wild-type cells under different inhibitor selections were controlled to be identical (two doublings between every time point), so even though raw Rosace scores are computed independently per condition, $\beta_{v,inh}$ are directly comparable between inhibitor conditions.

Within one condition (DMSO or inhibitor), according to convention, we call variants with $\beta_{v,inh} \gg 0$ or $\beta_{v,DMSO} \gg 0$ “gain-of-function” and $\beta_{v,inh} \ll 0$ or $\beta_{v,DMSO} \ll 0$ “loss-of-function”. With scores from multiple conditions, we presented three types of filtering strategies and produced the following classification: inhibitor-specific “resistance mutation”, inhibitor-specific “resistance position”, and “loss-of-function” and “gain-of-function” mutation in the context of growth rate differential with and without an inhibitor.

We stress the different interpretations of “gain-of-function” and “loss-of-function” labels. Within one condition, this label is a general term to describe whether the function of protein is perturbed by the mutation. In contrast, the latter describes the difference with and without a given inhibitor, canceling effects of folding, expression, and stability and targeting only the inhibitor sensitivity function of the protein.

A “resistance mutation” is specific to a certain inhibitor, and it satisfies the chained inequality $\beta_{v,inh} \gg \beta_{wt,inh} = \beta_{wt,DMSO} \beta_{v,DMSO}$. The first inequality specifies that the growth rate of a resistant mutation needs to be much larger than that of the wild-type in the presence of the inhibitor, and we used the onesided statistical test $\beta_{v,inh} > 0.5$ with the test statistics cutoff 0.1. The second

inequality specifies that in DMSO, the growth rate of that mutation is equal to or lower than that of the wild-type, ensuring that the resistance behavior we see is specific to that inhibitor, not that it grows faster under every condition, and thus we used the effect size cutoff $\beta_{v,DMSO} \leq 0$.

A “resistance position” is a position that contains at least one “resistance mutation” to a certain inhibitor.

In the context of growth rate differential with and without an inhibitor, a mutation is “gain-of-function” if it has a higher growth rate in the presence of the inhibitor than in its absence, which is one feature of “resistance mutation”. It is “loss-of-function” if it grows faster in the absence of the inhibitor. To label the mutations accordingly, we first computed a recentered Rosace score for each mutation under inhibitor selection $\gamma_{v,inh} = \beta_{v,inh} - \beta_{v,DMSO}$, and define “gain-of-function” $\gamma_{v,inh} > 0.75$ and “loss-of-function” $\gamma_{v,inh} < 0$ in the differential sensitivity analysis.

Machine learning modeling

Feature selection for the machine learning model

Interpretable features of the MET sequence variants and inhibitors were carefully chosen to be incrementally added to a model. To extract structural features from inhibitor bound mutant complexes, we used Umol to predict the structures of all the MET kinase variants bound to each of the inhibitors (Bryant et al. 2023 [\[1\]](#)). The input to Umol is the MET kinase variant sequence, SMILES string of the inhibitor and list of residues lining the putative binding pocket. The predicted complexes (MET kinase bound to inhibitor) were relaxed using OpenMM (Eastman et al. 2013 [\[2\]](#)). To ensure the inhibitor in the predicted structures are in the same pose as compared to reference structures, we tethered the predicted inhibitor structure to the reference pose using a modified version of the script available in https://github.com/Discngine/rdkit_tethered_minimization [\[3\]](#). The reference pose for crizotinib, NVP-BVU972, Merestenib and Savolitinib were taken from the corresponding crystal structures in the PDB - 2WGJ, 3QTI, 4EEV and 6SDE, while the reference pose for cabozantinib, capmantinib, glumetinib and glesatinib analog were taken from the structures docked using Autodock Vina (see Kinase domain structural analysis). Following this, the tethered inhibitors were redocked back to the predicted variant structures using Autodock Vina (Eberhardt et al. 2021 [\[4\]](#)). We also extracted features from wild-type MET kinase structures. The features could be broadly classified into four categories: inhibitor, stability, distance, conformation and inhibitor binding. Apart from these, ESM Log Likelihood Ratio was used as a feature in all models that we trained. Each of the feature categories that we explored and the rationale behind choosing them are explained below:

(1) ESM Log Likelihood Ratio (ESM LLR)

ESM1b is an unsupervised protein language model trained on a large set of protein sequences from UniProt that has successfully learned protein fitness patterns (Rives et al, 2021 [\[5\]](#); Lin et al., 2023 [\[6\]](#)). By including a mask token at a given position in the sequence, the log-likelihoods of all amino acid substitutions can be extracted from the model. The ratio between ESM1b log-likelihoods for the mutant and wildtype amino acids provides a score that indicates the fitness of each variant in the mutational scan, with log-likelihood ratios having precedent as a variant predictor (Rives et al, 2021 [\[5\]](#); Lin et al., 2023 [\[6\]](#)). The predictions used here were obtained using esm-variants webserver (https://huggingface.co/spaces/ntranoslab/esm_variants [\[7\]](#)) (Brandes et al., 2023 [\[8\]](#)).

(2) Inhibitor features

- **Inhibitor molecular weight:** We calculated the molecular weight of each inhibitor as a feature.

- **Ligand RMSD:** We structurally superposed the predicted variant structure onto the corresponding wildtype structure and calculated the RMSD between the predicted, redocked inhibitor and the reference inhibitor structure (**Figure 8 - figure supplement 1J** [↗](#))

(3) Stability features

- **$\Delta\Delta\Delta G$ and $\Delta\Delta G$:** Because inhibitor types are largely distinguished based on binding configuration, we reasoned that the difference in stability contributed by each mutation between given binding states (e.g. Type I bound state vs. a Type II bound state) could contribute to the success of the predictor. To compute the stability difference, we used structural representatives for type-I bound (2WGJ) and type II bound (4EEV) MET kinase and calculated the change in free energy ($\Delta\Delta G$) of every possible mutation at every position using ThermoMPNN (Dieckhaus et al., 2023 [↗](#)). The difference in $\Delta\Delta G$ between type-I bound and type-II bound structures ($\Delta\Delta\Delta G$) for every variant was added as a feature to the XGBoost model to capture the difference in stabilization from the mutation in the Type I or Type II bound state (**Figure 8 - figure supplement 1C** [↗](#)). We also used the predicted $\Delta\Delta G$ score of the corresponding inhibitor type-bound structure directly as a feature. For instance, if the input data corresponds to a mutation to Alanine at position 1065 in the presence of glumetinib (a type I inhibitor), the difference between $\Delta\Delta G$ predicted for the 1065A variant for the type-I bound (2WGJ) and for type II bound (4EEV) structure is used as a feature ($\Delta\Delta\Delta G$). The $\Delta\Delta G$ predicted for the 1065A variant for the type-I bound (2WGJ) structure is also used as a feature.

(4) Distance features

- **Residue to ATP distance:** Proximity to the ATP-binding site indicates the ability of the given residue to influence inhibitor binding given that Type I and Type II inhibitors are ATP competitive. To include this feature, the distance between C-alpha residue atoms and the centroid of bound ATP in a representative structure (PDB 3DKC) was calculated and the distance corresponding to each position was added as a feature (**Figure 8 - figure supplement 1D** [↗](#)).
- **Inhibitor distance:** This is the shortest distance between the inhibitor and mutated residue in the predicted variant-inhibitor complexes. (**Figure 8 - figure supplement 1G** [↗](#)).

(5) Conformational features

- **MET crystal structure RMSF:** The extent of flexibility at the mutation position could be significantly affected by the mutation, which in turn can affect the function of the variant. To account for this, we utilized the structural information abundantly available for MET kinases in PDB. We structurally aligned all crystal structures of human MET kinases with resolution better than 3 (81 structures) using mTM-align (Dong et al., 2018 [↗](#)) and calculated the Root Mean Squared Fluctuation at every residue position using Prody (Zhang et al., 2021 [↗](#)) (**Figure 8 - figure supplement 1F** [↗](#)).
- **Residue RMSD:** We structurally superposed the predicted variant structure onto the corresponding wildtype structure and calculated the RMSD between the mutant and wildtype residue at the mutation position (**Figure 8 - figure supplement 1I** [↗](#))

(6) Inhibitor Binding features

- **RF-Score:** To quantify the binding strength between the inhibitor and the variant protein structure, we calculated the RF-score, which is a random forest-based approach to predict protein-ligand binding affinity (Wójcikowski et al., 2017 [↗](#))

- **Pocket volume, hydrophobicity score and polarity score:** Changes to the binding pocket in terms of volume and hydrophobicity due to mutations could affect the interaction and binding between the inhibitor and variant. These effects were brought in as features into the model by calculating the binding pocket volume, hydrophobicity score and polarity score of the binding pocket using fpocket (Le Guilloux et al., 2009) (Figure 8 - figure supplement 1H).

This category of features are not part of the best performing model shown in Figure 8. Apart from these categories, we calculated the difference in volume between the wildtype and mutated residue at a given position and added it as a feature (Δ Volume) since residue volume changes upon mutation could contribute to steric hindrance (Figure 8 - figure supplement 1E). This feature is also not part of the best performing model.

This led to a total of 14 interpretable features to evaluate our models on. We trained and tested a total of 8192 models by considering all possible numbers and combinations of these features (keeping ESM LLR as a constant feature in all models). The hyperparameter tuning, cross-validation, training and testing of each of these models are described in detail below.

Training and Selecting the Predictive Model

An XGBoost regressor model, which is a gradient boosting method based on decision trees as the base learner (Chen and Guestrin, 2016), was used to predict DMS fitness scores in presence of inhibitors. Given the relatively small dataset we are using here, the models are prone to overfitting. Hence, we used monotonic constraints on features that had a monotonic relationship with the experimental fitness scores. ESM LLR score and $\Delta\Delta G$ have a positive and negative correlation with the experimental fitness scores respectively (Figure 8 - figure supplement 1A). Therefore, ESM LLR was constrained positively and $\Delta\Delta G$ was constrained negatively by assigning 1 and -1 respectively to the 'monotone_constraints' parameter in Python XGBoost. This ensures that the monotonic relationship between the input feature and the target value is maintained during predictions. To further prevent overfitting, we binned the values of the 12 remaining into four or five bins and assigned the median of the bin as their value. The bins were chosen such that one or two bins would contain the majority of feature values. The distribution of these twelve features are shown in Figure 8 - figure supplement 1B. The bins of each feature are shown as red dashed lines on the histograms. Model performance was evaluated using Pearson's R and mean squared error (MSE).

Experimental fitness scores of MET variants in the presence of DMSO and AMG458 were ignored in model training and testing since having just one set of data for a type I $\frac{1}{2}$ inhibitor and DMSO leads to learning by simply memorizing the inhibitor type, without generalizability. The remaining dataset was split into training and test sets to further avoid overfitting (Figure 8A). The following data points were held out for testing - (a) all mutations in the presence of one type I (crizotinib) and one type II (glesatinib analog) inhibitor, (b) 20% of randomly chosen positions (columns) and (c) all mutations in two randomly selected amino acids (rows) (e.g. all mutations to Phe, Ser). After splitting the dataset into train and test sets, the train set was used for XGBoost hyperparameter tuning and cross-validation. For tuning the hyperparameters of each of the XGBoost models, we held out 20% of randomly sampled data points in the training set and used the remaining 80% data for Bayesian hyperparameter optimization of the models with Optuna (Akiba et al., 2019), with an objective to minimize the mean squared error between the fitness predictions on 20% held out split and the corresponding experimental fitness scores. The following hyperparameters were sampled and tuned: type of booster (booster - gbtrees or dart), maximum tree depth (max_depth), number of trees (n_estimators), learning rate (eta), minimum leaf split loss (gamma), subsample ratio of columns when constructing each tree (colsample_bytree), L1 and L2 regularization terms (alpha and beta) and tree growth policy (grow_policy - depthwise or lossguide). After identifying the best combination of hyperparameters for each of the models, we

performed 10-fold cross validation (with re-sampling) of the models on the full training set. The training set consists of data points corresponding to 230 positions and 18 amino acids. We split these into 10 parts such that each part corresponds to data from 23 positions and 2 amino acids. Then, at each of 10 iterations of cross-validation, models were trained on 9 of 10 parts (207 positions and 16 amino acids) and evaluated on the 1 held out part (23 positions and 2 amino acids). Through this protocol we ensure that we evaluate performance of the models with different subsets of positions and amino acids. The average Pearson correlation and mean squared error of the models from these 10 iterations were calculated and the best performing model out of 8192 models was chosen as the one with the highest cross-validation correlation. The final XGBoost models were obtained by training on the full training set and also used to obtain the fitness score predictions for the validation and test sets. These predictions were used to calculate the inhibitor-wise correlations shown in **Figure 8B** [↗](#).

Kinase domain structural analysis

Unless otherwise stated, all structural analysis was performed on PyMOL. Structural mapping incorporated tools from the Bio3D bioinformatics package in R ([Grant et al., 2006](#) [↗](#)). Inhibitors that lacked an experimental crystal structure were docked into a representative type I (2WGJ) or type II (4EEV) structure with AutoDock Vina ([Eberhardt et al., 2021](#) [↗](#)). Existing ligands in both the structures were removed *in silico* and the proteins prepared for docking using AutoDockTools by adding polar hydrogens and Kollman charges. The inhibitors were also prepared using AutoDockTools by adding polar hydrogens and charges and identifying rotatable torsions. A grid box which dictates the search space for the docking tool was defined approximately around the region where the existing ligands in 2WGJ and 4EEV were bound. The energy range and exhaustiveness of docking was set to 3 and 8 respectively. AutoDock Vina was made to output 5 modes for each ligand. Capmatenib and glumetinib (type I inhibitors) were docked on to 2WGJ and glesatinib analog and cabozantinib (type II inhibitors) were docked on to 4EEV.

Supplemental information

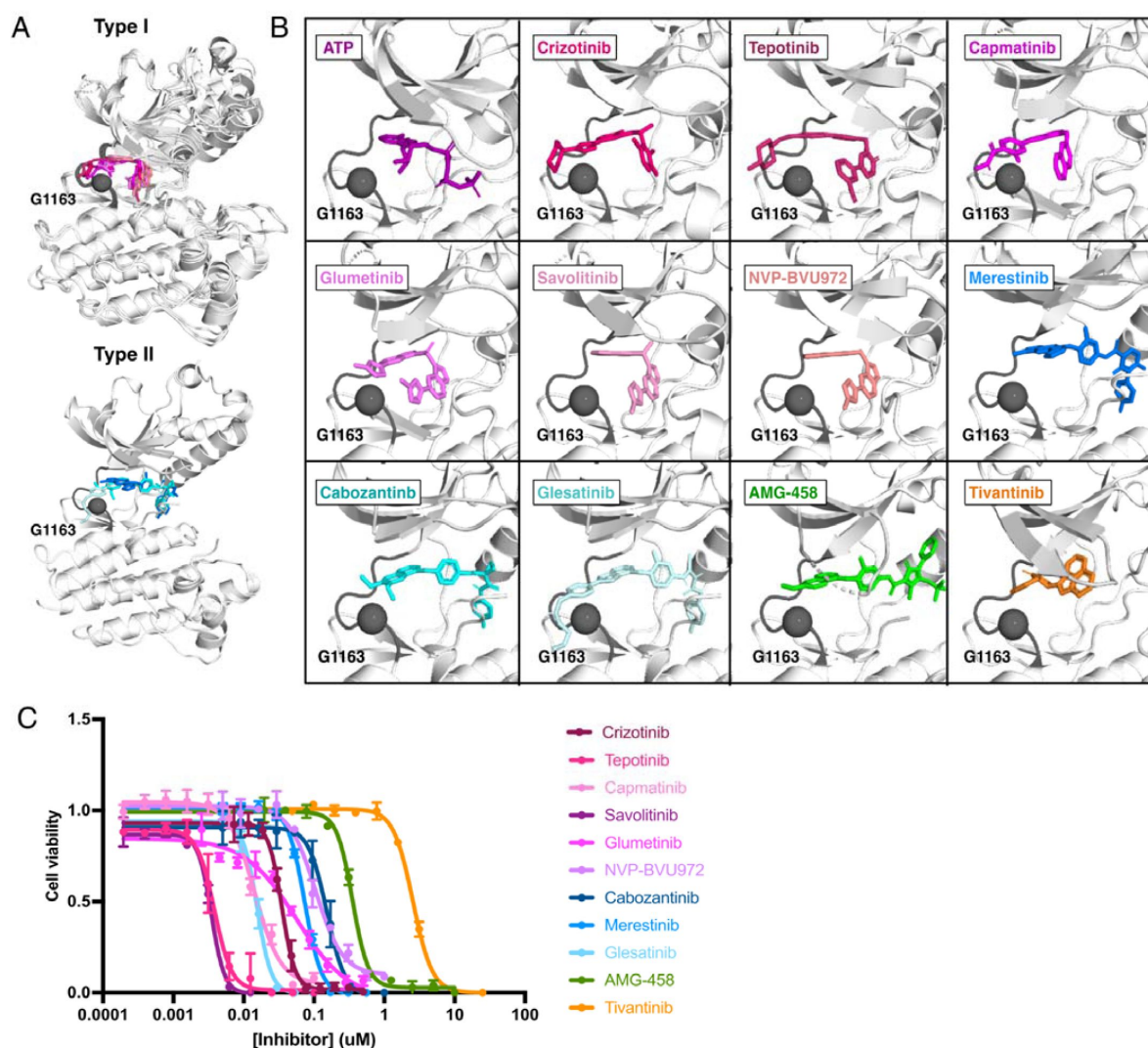


Figure 1 - figure supplement 1.

Structural inhibitor classification and dose-response determination.

(A) Type I (crizotinib, 2WGJ; tepotinib, 4R1V; capmatinib; savolitinib, 6SDE; NVP-BVU972, 3QTI) and type II (merestinib, 4EEV; cabozantinib; glesatinib) inhibitor-bound MET kinase domain structures globally aligned. Hinge (gray) and G1163 (represented as a sphere) are highlighted to show the kinase domain solvent-front relative to each inhibitor. Inhibitors lacking experimental structures (capmatinib, cabozantinib, glumetinib, and glesatinib) were docked onto a representative type I (PDB 2WGJ) and type II (4EEV) structure through AutoDock Vina (Eberheart et al., 2021; Trott et al., 2010). (B) Solvent-front and G1163 highlighted relative to the ATP-bound kinase domain crystal structure (3DKC) and all inhibitors screened. (C) Dose-response curves for each inhibitor against the TPR-fusion MET and METΔEx14 intracellular domains, stably expressed in Ba/F3 cells.

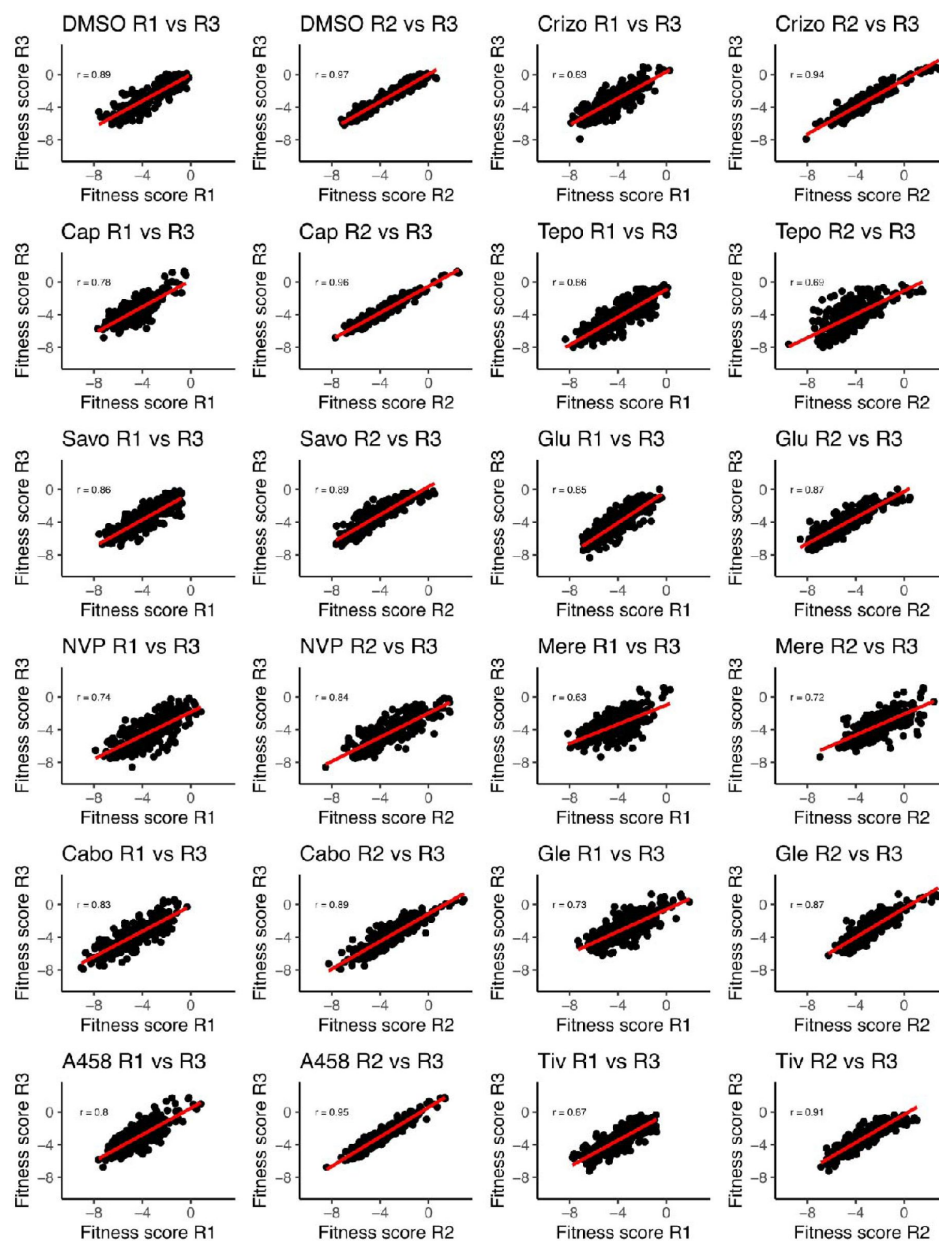


Figure 1 - figure supplement 2.

Correlation analysis of the MET kinase domain site saturation mutagenesis library across replicates and conditions.

Replicate correlation analysis for each inhibitor for both the TPR-fusion MET background scores with Enrich2.

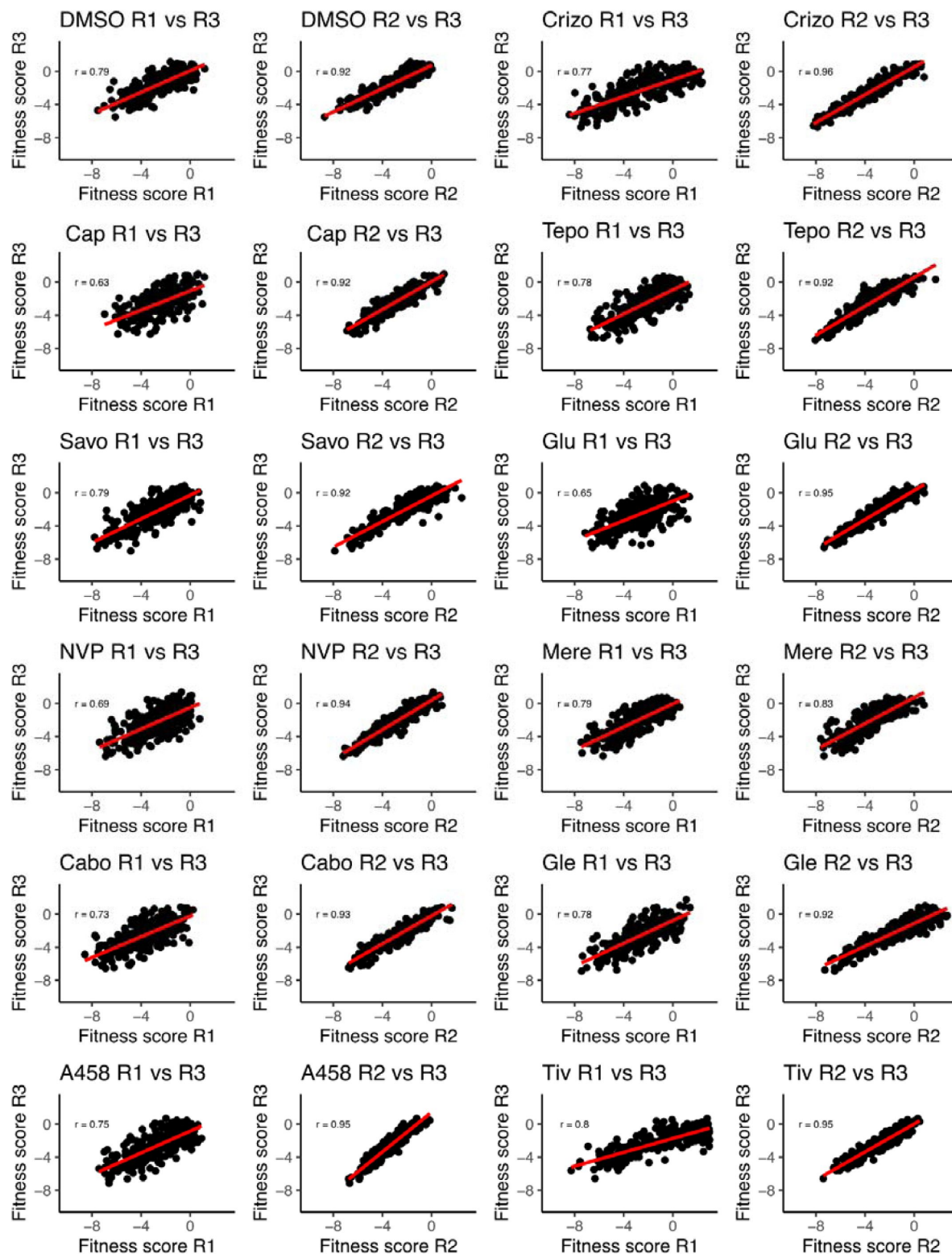


Figure 1 - figure supplement 3.

Correlation analysis of the METΔEx14 kinase domain site saturation mutagenesis library across replicates and conditions.

Replicate correlation analysis for each inhibitor for the TPR-fusion METΔEx14 background score with Enrich2.

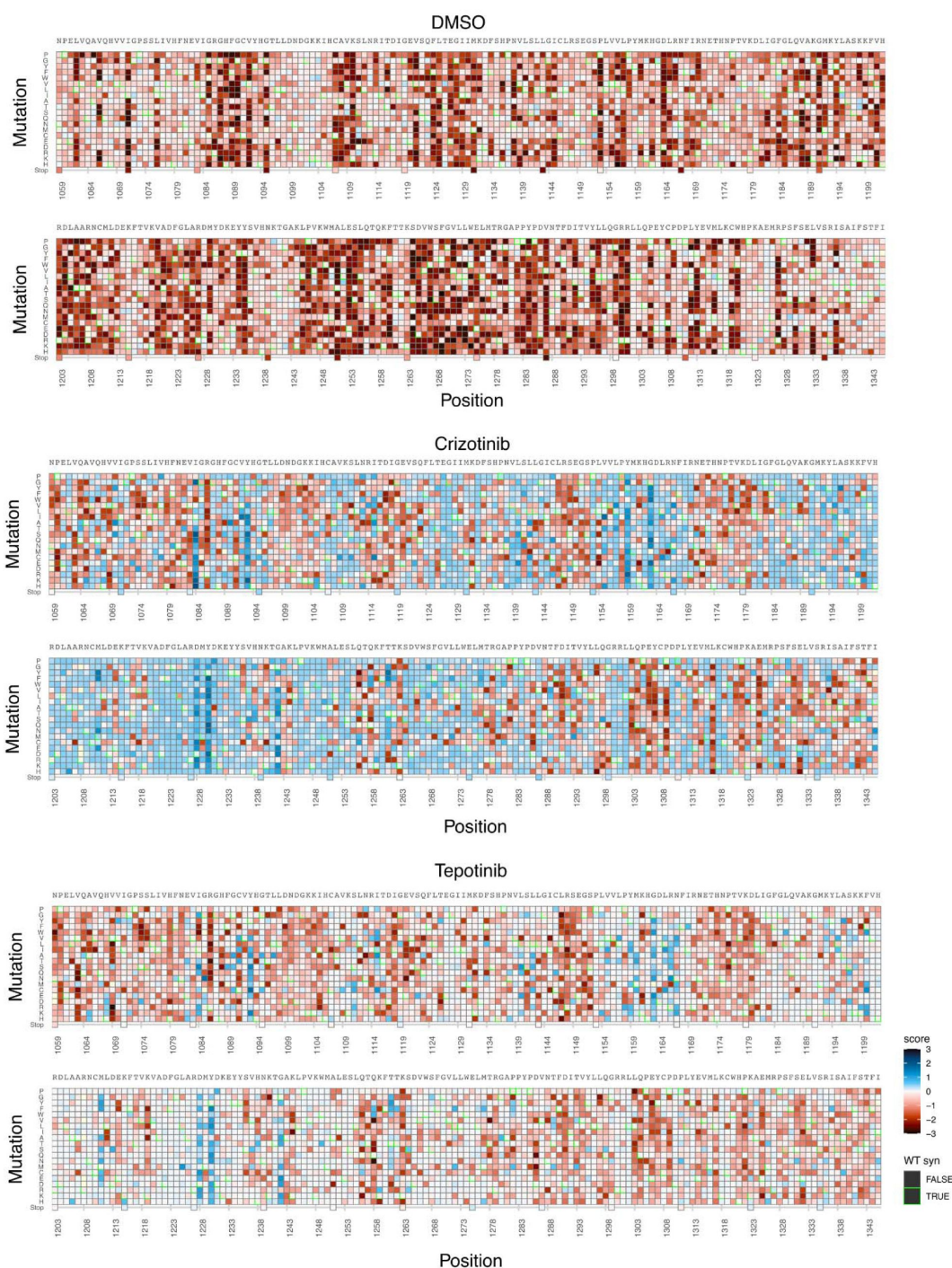


Figure 1 - figure supplement 4.

Fitness landscapes of the MET kinase domain against a panel of 11 inhibitors.

Heatmap for the DMSO control condition and all inhibitor fitness scores from Rosace, subtracted from DMSO for >99% of MET kinase domain variants in the full intracellular domain background in the context of the TPR-fusion. Wild-type synonymous mutations are highlighted in green, and mutations that were not captured by the screen are in light yellow.

MET kinase domain heatmaps (2/4)

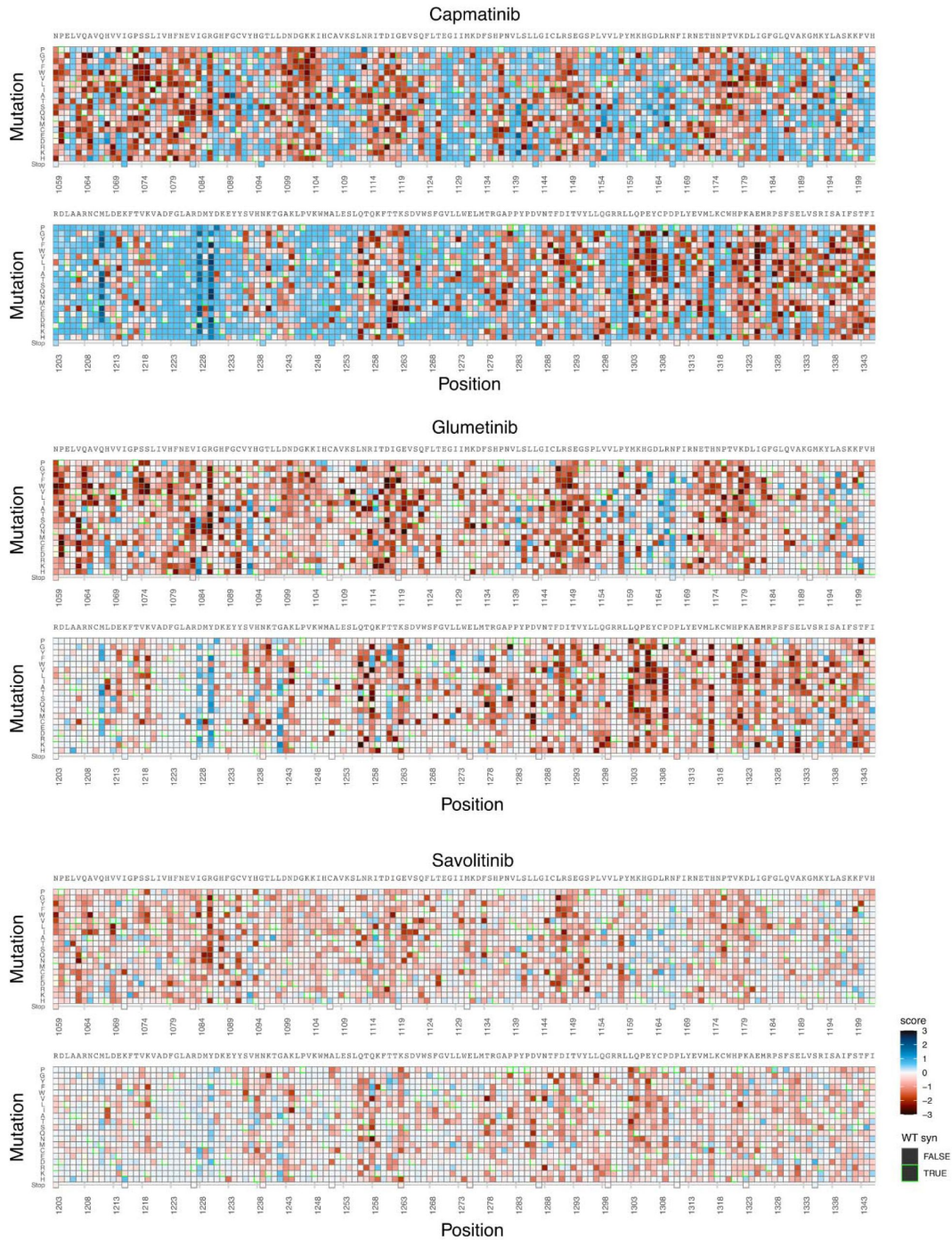


Figure 1 - figure supplement 4. (continued)

MET kinase domain heatmaps (3/4)

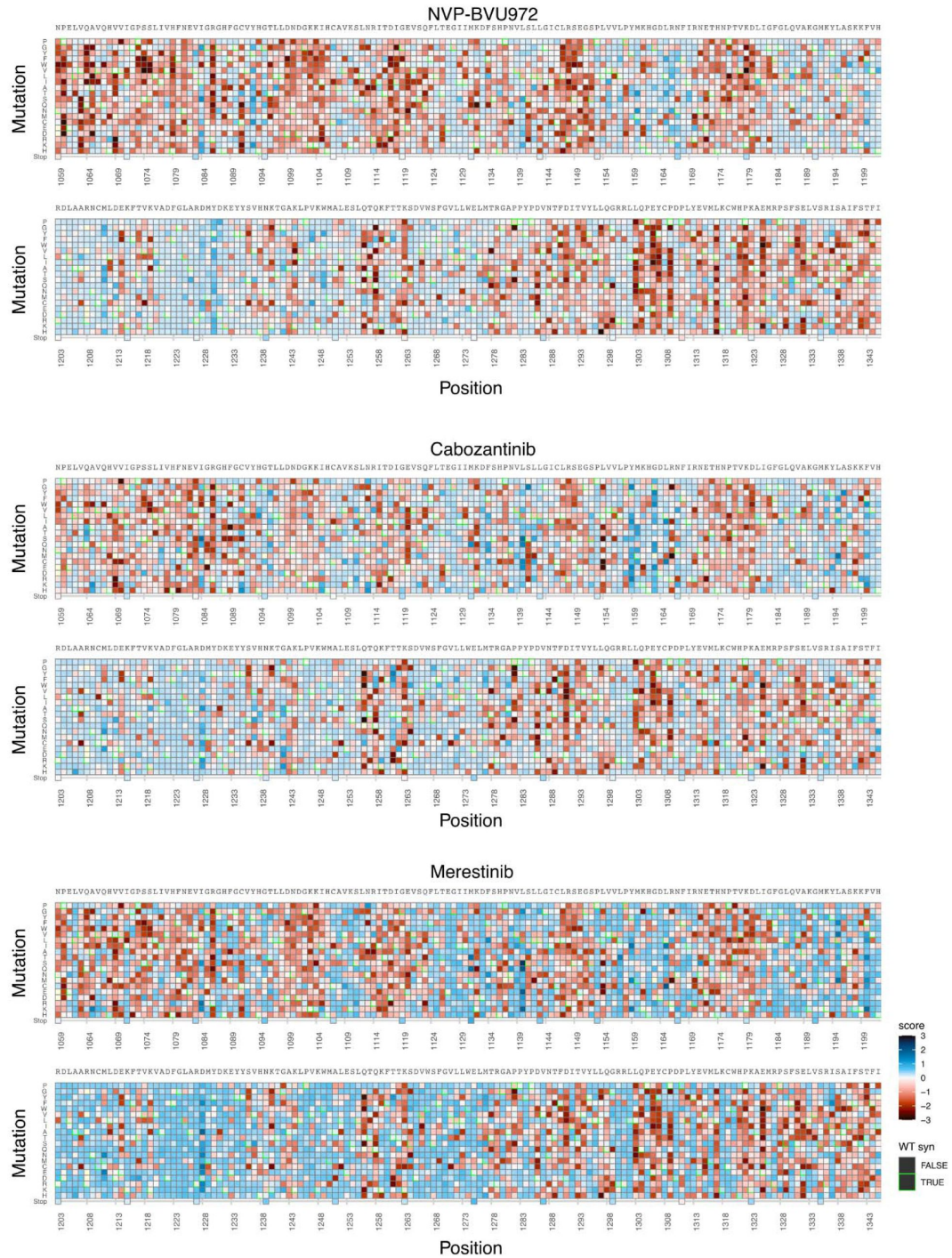


Figure 1 - figure supplement 4. (continued)

MET kinase domain heatmaps (4/4)

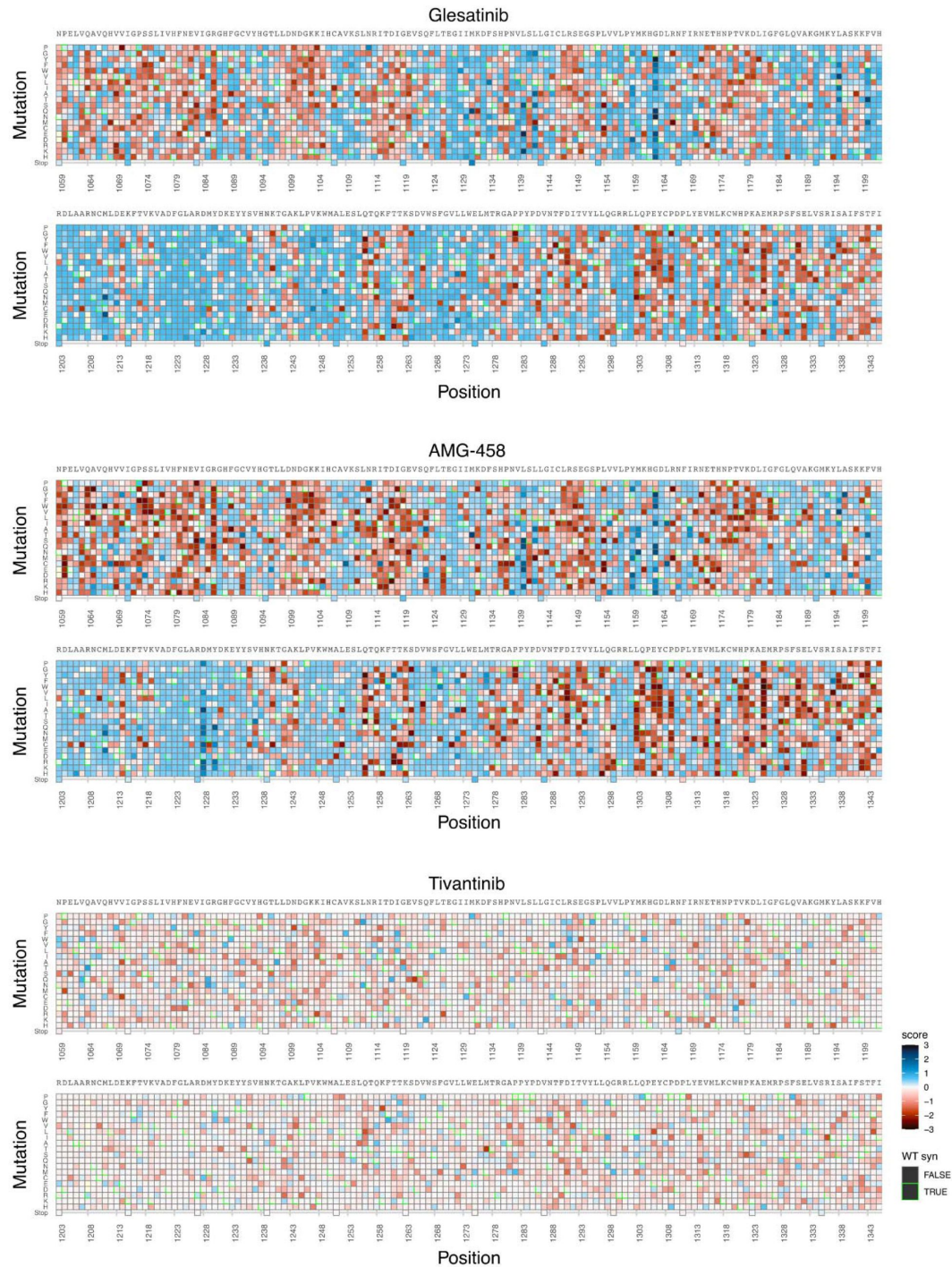


Figure 1 - figure supplement 4. (continued)

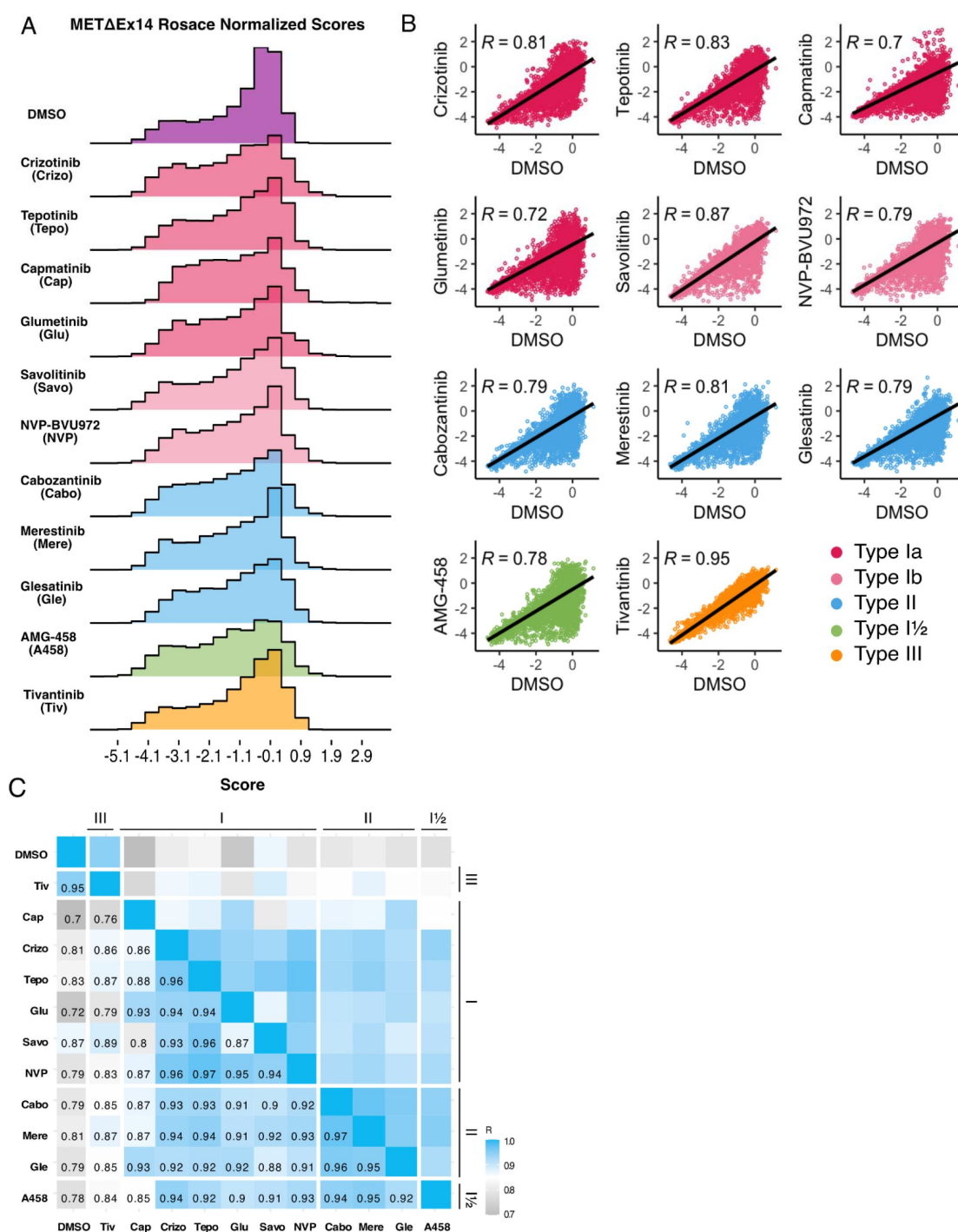


Figure 2 - figure supplement 5.

Mutational landscape of the METΔEx14 kinase domain under 11 ATP-competitive inhibitor selection.

(A) Distributions of all variants (**wild-type** synonymous, early stop, and missense) for each condition, scored with Rosace and normalized to the growth rate of the DMSO control population. (B) Correlation plots for all mutational fitness scores for each drug against DMSO, fitted with a linear regression and Pearson's R value displayed. (C) Heatmap showing the Pearson's R correlation for each condition against each other, annotated by condition and inhibitor type. Correlations are colored according to a scale bar from gray to blue (low to high correlation).

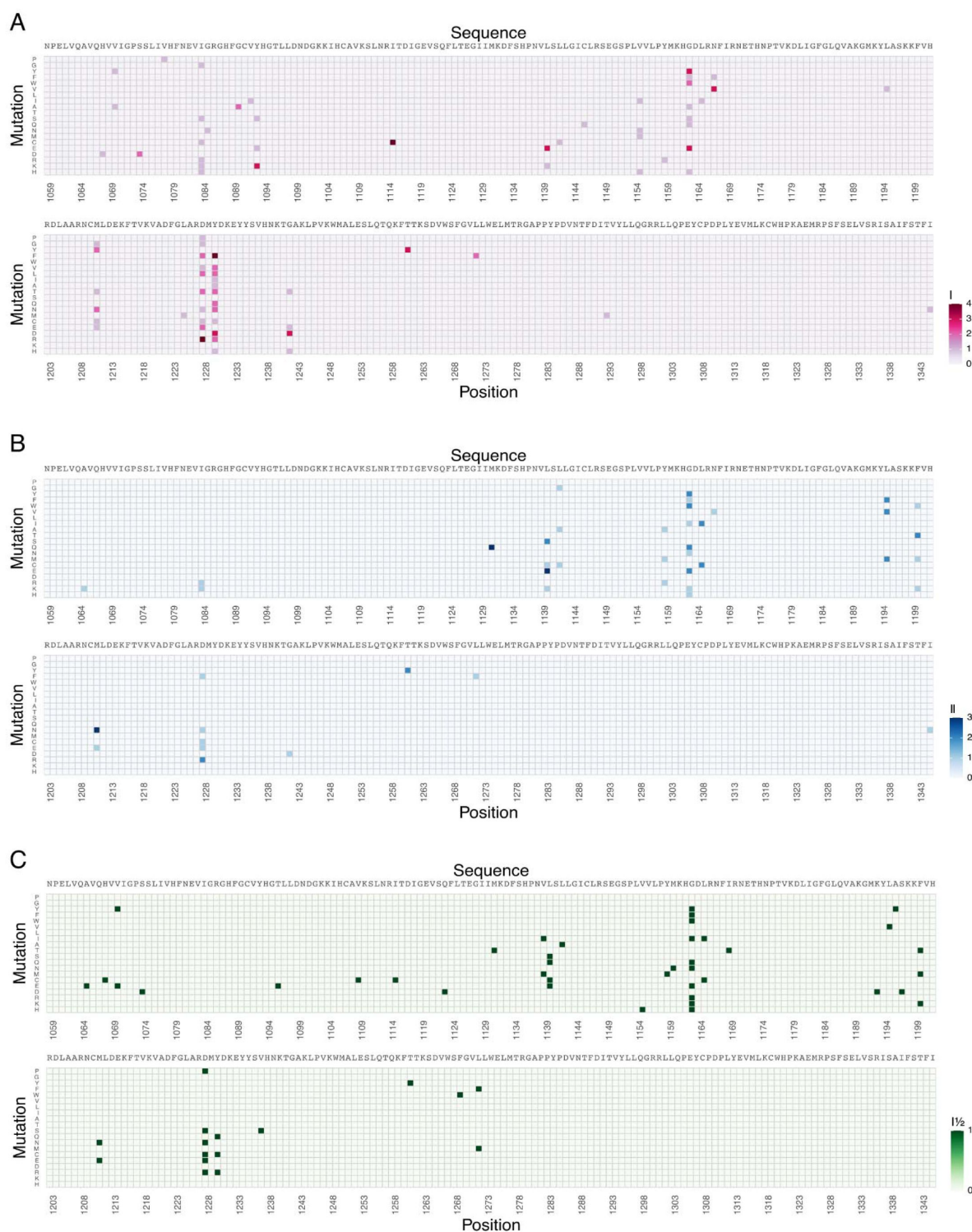


Figure 6 - figure supplement 1.

Statistically filtered resistance mutations for grouped type I, type II, and type I½ inhibitors for MET.

(A-C) Heatmaps of the sum of resistance mutations grouped for type I (pink), type II (blue), and type I½ (green) for MET.

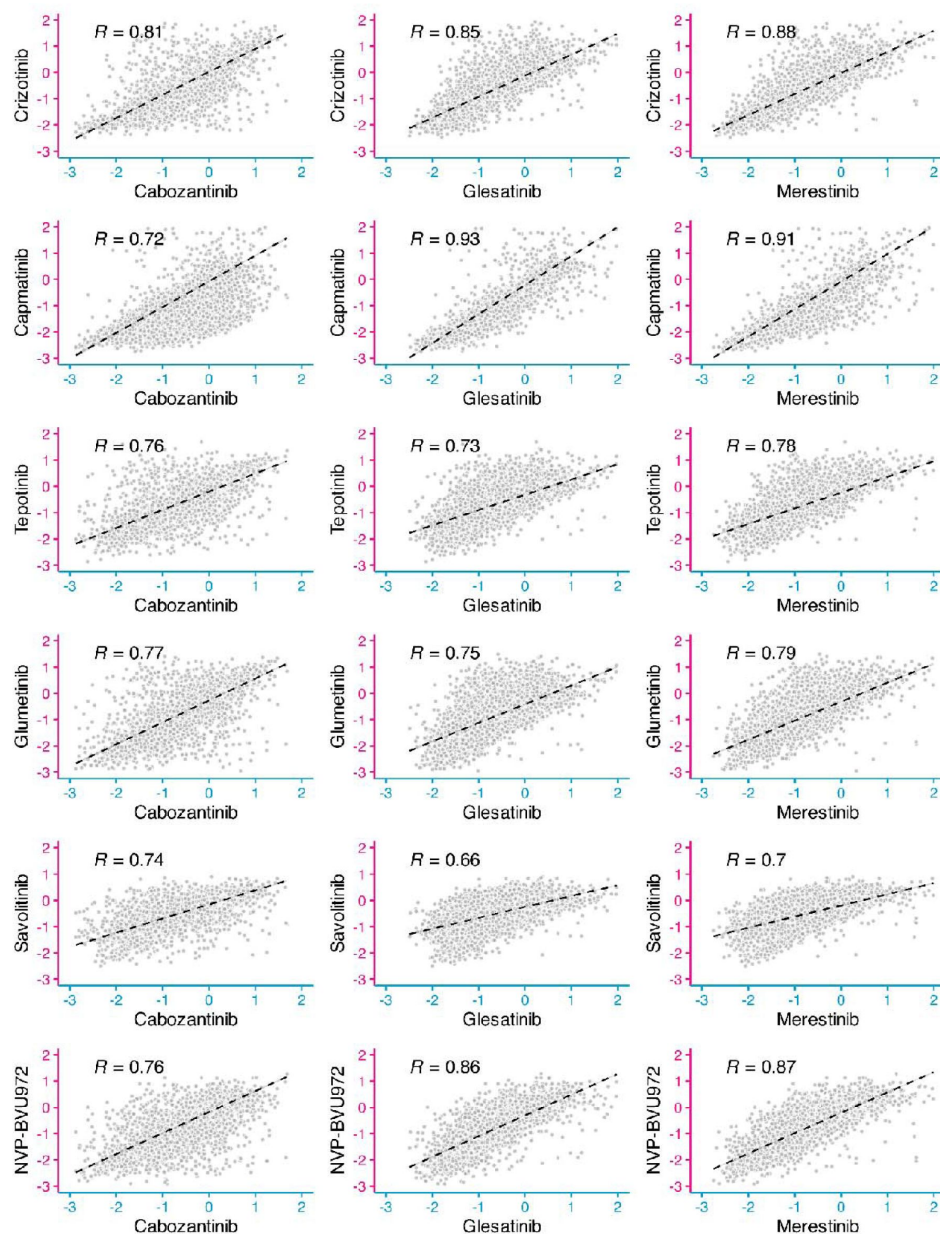


Figure 7 - figure supplement 1.

Cross-comparison of type I and type II inhibitor pairs.

Scatter plots of each type II inhibitor fitness scores (cabozantinib, glesatinib analog, merestinib; axis in blue) against each type I inhibitor (crizotinib, capmatinib, tepotinib, glumetinib, savolitinib, NVP-BVU972; axis in pink).

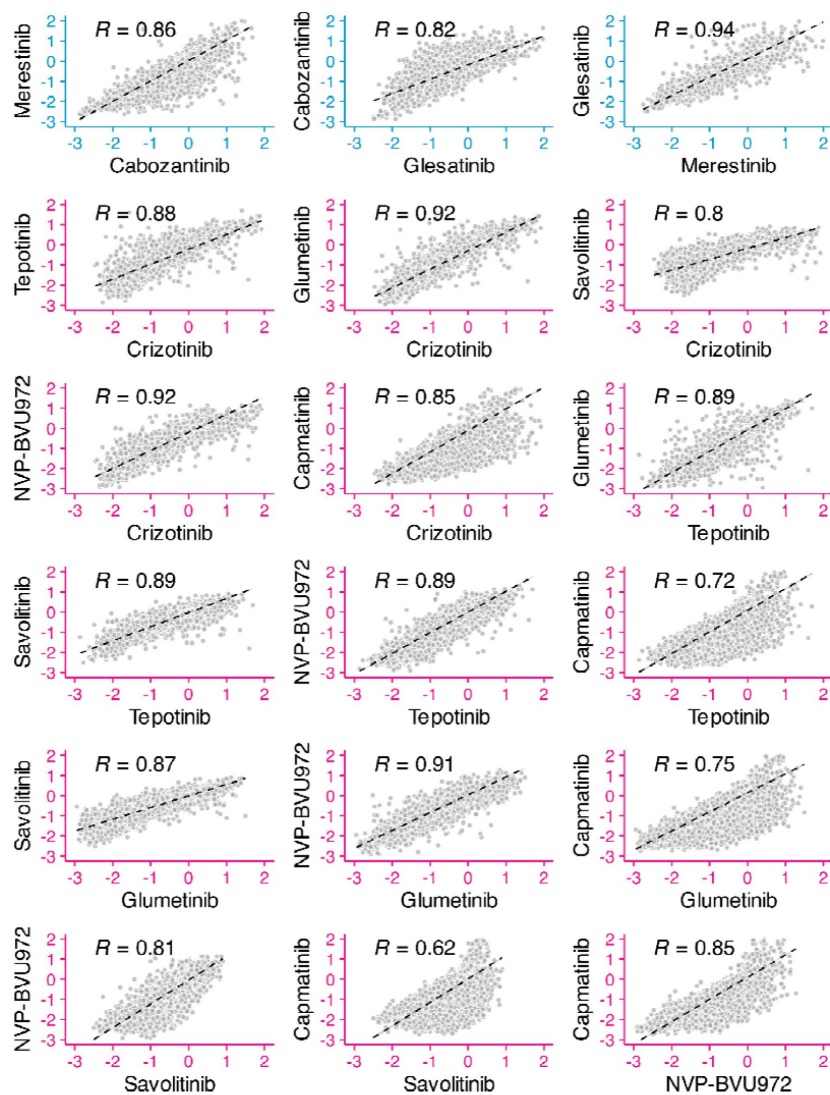


Figure 7 - figure supplement 2.

Cross-comparison analysis of inhibitors within the same type.

Scatter plots of each inhibitor pair within the type II group (cabozantinib, glesatinib analog, merestinib; axes in blue) and within the type I group (crizotinib, capmatinib, tepotinib, glumetinib, savolitinib, NVP-BVU972; axes in pink).

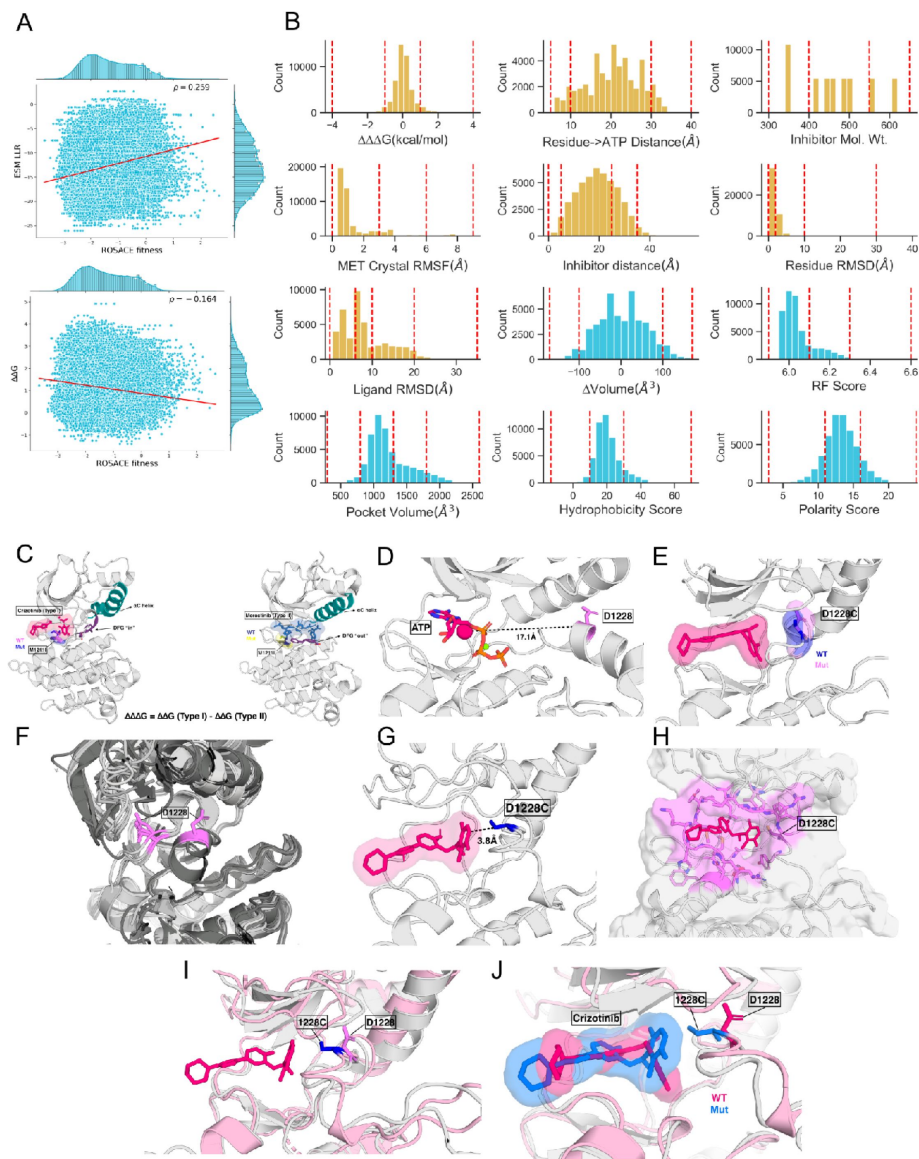


Figure 8 - figure supplement 1.

Distribution and visualization of features used in the XGBoost machine learning models.

(A) Distribution of ESM LLR vs. Experimental fitness (top) and $\Delta\Delta G$ vs. Experimental fitness (bottom). (B) Distribution of all features (except ESM LLR and $\Delta\Delta G$) extracted and used for the XGBoost models. The features that were incorporated in the best performing model are shown in yellow. The red dashed lines within each distribution show the edges of bins used to bin the feature values. (C) $\Delta\Delta G$ calculated from predicted $\Delta\Delta G$ Type I (PDB 2WGJ) (left) and Type II (PDB 4EEV) (right) MET kinase structure by subtracting type II $\Delta\Delta G$ from type I $\Delta\Delta G$. The key regions showing difference in conformation between type I and II structures are the DFG motif (purple) and αC helix (teal). (D) Calculation of "residue to ATP" distance feature for residue D1228 in ATP bound MET Kinase structure (3DKC) is shown. Centroid of the ATP molecule is shown as a pink sphere. (E) Example of $\Delta Volume$ feature calculation using the difference between the volume of Asp and Cys. (F) Ensemble of MET kinase domain crystal structures aligned and RMSF of a given residue (D1228 in this example). (G) The shortest distance between the inhibitor and a mutation calculated from the Umol predicted variant-inhibitor structure. (H) The binding pocket of crizotinib in the predicted Umol structure. Pocket volume, hydrophobicity score, polarity score and RF score are calculated from this binding site. (I) Residue RMSD feature is described by the Umol predicted structure of variant D1228C (pink) superposed onto the wild-type reference structure (PDB 2WGJ, gray) and RMSD between D1228 in the wild-type structure and D1228C in the variant structure. (J) Ligand RMSD feature The Umol predicted structure of variant D1228C (pink) superposed onto the wild-type reference structure (PDB 2WGJ, gray) and RMSD between crizotinib in the wild-type structure (pink) and in the variant structure (blue).

Code and data availability

The sequencing data has been deposited at the NCBI SRA (BioProject PRJNA1136906). Original data files, analysis, and source code is available at https://github.com/fraser-lab/MET_kinase_Inhibitor_DMS 

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Additional information

Author contributions

Conceptualization JSF, NJ, EAC; Methodology, GOE, EML, JR, CBM, AR, KC; Formal analysis, GOE, AR; Writing – Original Draft, GOE and JSF; Writing – Editing, GOE, JSF, EML, NJ, EAC; Validation, GOE, EML,; Funding Acquisition, JSF, NJ, EAC; Resources, JSF, NJ, EA, WCM, HP; Supervision, JSF, NJ, EAC, WCM, EML, HP, JAC.

Additional files

Response to reviews 

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Reviewer #2 (Public review):**Summary:**

This manuscript provides a comprehensive overview of potential resistance mutations within MET Receptor Tyrosine Kinase and defines how specific mutations affect different inhibitors and modes of target engagement. The goal is to identify inhibitor combinations with the lowest overlap in their sensitivity to resistant mutations and determine if certain resistance mutations/mechanisms are more prevalent for specific modes of ATP-binding site engagement. To achieve this, the authors measured the ability of ~6000 single mutants of MET's kinase domain (in the context of a cytosolic TPR fusion) to drive IL-3-independent proliferation (used as a proxy for activity) of Ba/F3 cells (deep mutational profiling) in the presence of 11 different inhibitors. The authors then used co-crystal and docked structures of inhibitor-bound MET complexes to define the mechanistic basis of resistance and applied a protein language model to develop a predictive model of inhibitor sensitivity/resistance.

Strengths:

The major strengths of this manuscript are the comprehensive nature of the study and the rigorous methods used to measure the sensitivity of ~6000 MET mutants in a pooled format. The dataset generated will be a valuable resource for researchers interested in understanding kinase inhibitor sensitivity and, more broadly, small molecule ligand/protein interactions. The structural analyses are systematic and comprehensive, providing interesting insights into resistance mechanisms. Furthermore, the use of machine learning to define inhibitor-specific fitness landscapes is a valuable addition to the narrative. Although the ESM1b protein language model is only moderately successful in identifying the underlying mechanistic basis of resistance, the authors' attempt to integrate systematic sequence/function datasets with machine learning serves as a foundation for future efforts.

Weaknesses:

The main limitation of this study is that the authors' efforts to define general mechanisms between inhibitor classes were only moderately successful due to the challenge of uncoupling inhibitor-specific interaction effects from more general mechanisms related to the mode of ATP-binding site engagement. However, this is a minor limitation that only minimally detracts from the impressive overall scope of the study.

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Reviewer #3 (Public review):**Summary:**

In the manuscript 'Mapping kinase domain resistance mechanisms for the MET receptor tyrosine kinase via deep mutational scanning' by Estevam et al, deep mutational scanning is used to assess the impact of ~5,764 mutants in the MET kinase domain on the binding of 11 inhibitors. Analyses were divided by individual inhibitor and kinase inhibitor subtype (I,II, I 1/2, and III). While a number of mutants were consistent with previous clinical reports, novel potential resistance mutants were also described. This study has implications for the development of combination therapies, namely which combination of inhibitors to avoid based on overlapping resistance mutant profiles. While one suggested pair of inhibitors with least overlapping resistance mutation profiles was suggested, this manuscript presents a proof of concept toward a more systematic approach for improved selection of combination therapeutics. Furthermore, in a final part of this manuscript the data was used to train a machine learning model, the ESM-1b protein language model augmented with an XG Boost

Regressor framework, and found that they could improve predictions of resistance mutations above the initial ESM-1b model.

Strengths:

Overall this paper is a tour-de-force of data collection and analysis to establish a more systematic approach for the design of combination therapies, especially in targeting MET and other kinases, a family of proteins significant to therapeutic intervention for a variety of diseases. The presentation of the work is mostly concise and clear with thousands of data points presented neatly and clearly. The discovery of novel resistance mutants for individual MET inhibitors, kinase inhibitor subtypes within the context of MET, and all resistance mutants across inhibitor subtypes for MET has clinical relevance. However, probably the most promising outcome of this paper is the proposal of the inhibitor combination of Crizotinib and Cabozantinib as Type I and Type II inhibitors, respectively, with the least overlapping resistance mutation profiles and therefore potentially the most successful combination therapy for MET. While this specific combination is not necessarily the point, it illustrates a compelling systematic approach for deciding how to proceed in developing combination therapy schedules for kinases. In an insightful final section of this paper, the authors approach using their data to train a machine learning model, perhaps understanding that performing these experiments for every kinase for every inhibitor could be prohibitive to applying this method in practice.

Weaknesses:

This paper presents a clear set of experiments with a compelling justification. The content of the paper is overall of high quality. Below are mostly regarding clarifications in presentation.

Two places could use more computational experiments and analysis, however. Both are presented as suggestions, but at least a discussion of these topics would improve the overall relevance of this work. In the first case it seems that while the analyses conducted on this dataset were chosen with care to be the most relevant to human health, further analyses of these results and their implications of our understanding of allosteric interactions and their effects on inhibitor binding would be a relevant addition. For example, for any given residue type found to be a resistance mutant are there consistent amino acid mutations to which a large or small or effect is found. For example is a mutation from alanine to phenylalanine always deleterious, though one can assume the exact location of a residue matters significantly. Some of this analysis is done in dividing resistance mutants by those that are near the inhibitor binding site and those that aren't, but more of these types of analyses could help the reader understand the large amount of data presented here. A mention at least of the existing literature in this area and the lack or presence of trends would be worthwhile. For example, is there any correlation with a simpler metric like the Grantham score to predict effects of mutations (in a way the ESM-1b model is a better version of this, so this is somewhat implicitly discussed).

Indeed, this discussion relates to the second point this manuscript could improve upon: the machine learning section. The main actionable item here is that this results section seems the least polished and could do a better job describing what was done. In the figure it looks like results for certain inhibitors were held out as test data - was this all mutants for a single inhibitor, or some other scheme? Overall I think the implications of this section could be fleshed out, potentially with more experiments. As mentioned in the 'Strengths' section, one of the appealing aspects of this paper is indeed its potential wide applicability across kinases - could you use this ML model to predict resistance mutants for an entirely different kinase? This doesn't seem far-fetched, and would be an extremely compelling addition to this paper to prove the value of this approach.

Another area in which this paper could improve its clarity is in the description of caveats of the assay. The exact math used to define resistance mutants and its dependence on the DMSO control is interesting, it is worth discussing where the failure modes of this procedure might be. Could it be that the resistance mutants identified in this assay would differ significantly from those found in patients? That results here are consistent with those seen in the clinic is promising, but discrepancies could remain. Furthermore a more in depth discussion of the MetdelEx14 results is warranted. For example, why is the DMSO signature in Figure 1 - supplement 4 so different from that of Figure 1? And finally, there is a lot of emphasis put on the unexpected results of this assay for the tivantinib "type III" inhibitor - could this in fact be because the molecule "is highly selective for the inactive or unphosphorylated form of c-Met" according to Eathiraj et al JBC 2011? These points are addressed in previous work (Estevam et al 2024) or in the detailed methods section, but are not obvious in the main text of the paper.

This paper is crisply written with beautiful figures, and the complexity of the data is easy to understand from an in depth discussion of the mutants that have been previously reported.

Finally, the potential impacts and follow-ups of this excellent study could be used as a resource for the community both as a dataset and as a proof of concept. It is exciting that his approach can be altered and/or improved in the future to facilitate the general application of this approach for combination therapies and the understanding of mechanism for other targets.

Comments on revisions:

Thank you for your additions and changes - they have improved the quality of this paper.

<https://doi.org/10.7554/eLife.101882.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

In this work, the authors present a cornucopia of data generated using deep mutational scanning (DMS) of variants in MET kinase, a protein target implicated in many different forms of cancer. The authors conducted a heroic amount of deep mutational scanning, using computational structural models to augment the interpretation of their DMS findings.

Strengths:

This powerful combination of computational models, experimental structures in the literature, dose-response curves, and DMS enables them to identify resistance and sensitizing mutations in the MET kinase domain, as well as consider inhibitors in the context of the clinically relevant exon-14 deletion. They then try to use the existing language model ESM1b augmented by an XGBoost regressor to identify key biophysical drivers of fitness. The authors provide an incredible study that has a treasure trove of data on a clinically relevant target that will appeal to many.

We thank Reviewer 1 for their generous assessment of our manuscript!

Weaknesses:

However, the authors do not equally consider alternative possible mechanisms of resistance or sensitivity beyond the impact of mutation on binding, even though the measure used to discuss resistance and sensitivity is ultimately a resistance score derived from the increase or decrease of the presence of a variant during cell growth.

For this resistance screen, Ba/F3 was a carefully chosen cellular selection system due to its addiction to exogenously provided IL-3, undetected expression of endogenous RTKs (including MET), and dependence on kinase transgenes to promote signaling and growth under IL-3 withdrawal. Together this allows for the readout of variants that alter kinase-driven proliferation without the caveat of bypass resistance. In our previous phenotypic screen (Estevam et al., 2024, eLife), we also carefully examined the impact of all possible MET kinase domain mutations both in the presence and absence of IL-3 withdrawal, but no inhibitors. There, we identified a small group of mutations that were associated with gain-of-function behavior located at conserved regulatory motifs outside of the catalytic site, yet these mutations were largely sensitive to inhibitors within this screen.

Here, the majority of resistance mutations were located at or near the ATP-binding pocket, suggesting an impact on resistance through direct drug interactions. However, there was also a small population of distal mutations that met our statistical definitions of resistance. Within the crizotinib selection, sites such as T1293, L1272, T1261, amongst others, demonstrated resistance profiles but were located in C-lobe away from the catalytic site. While we did not experimentally validate these specific mutations, it is possible that non-direct drug binders instead promote resistance through allosteric or conformational mechanisms which preserve kinase activity and signaling. Indeed, our ML framework explicitly included conformational and stability effects as significant in improving predictions.

We would be happy to further discuss any specific alternative resistance mechanisms Reviewer 1 has in mind! Thank you for highlighting this!

There are also points of discussion and interpretation that rely heavily on docked models of kinase-inhibitor pairs without considering alternative binding modes or providing any validation of the docked pose. Lastly, the use of ESM1b is powerful but constrained heavily by the limited structural training data provided, which can lead to misleading interpretations without considering alternative conformations or poses.

The majority of our interpretations are grounded in the X-ray structures of WT MET bound to the inhibitors studied (or close analogs). The use of docked models (note - to mutant structures predicted by UMol, not ESM, that can have conformational changes) is primarily in the ML part of the manuscript. Indeed, in our models, conformational and binding mode changes are taken into account as features (see Ligand RMSD, Residue RMSD). There are certainly improved methods (AF3 variants) emerging that might have even more power to model these changes, but they come with greater computational costs and are something we will be evaluating in the future.

We added to the results section: “While our features can account for some changes in MET-mutant conformation and altered inhibitor binding pose, the prediction of these aspects can likely be improved with new methods.”

Reviewer #2 (Public review):

Summary:

This manuscript provides a comprehensive overview of potential resistance mutations within MET Receptor Tyrosine Kinase and defines how specific mutations affect different inhibitors and modes of target engagement. The goal is to identify inhibitor combinations with the lowest overlap in their sensitivity to resistant mutations and determine if certain resistance mutations/mechanisms are more prevalent for specific modes of ATP-binding site engagement. To achieve this, the authors measured the ability of ~6000 single mutants of MET's kinase domain (in the context of a cytosolic TPR fusion) to drive IL-3-independent proliferation (used as a proxy for activity) of Ba/F3 cells (deep mutational profiling) in the presence of 11 different inhibitors. The authors then used co-crystal and docked structures of inhibitor-bound MET complexes to define the mechanistic basis of resistance and applied a protein language model to develop a predictive model of inhibitor sensitivity/resistance.

Strengths:

The major strengths of this manuscript are the comprehensive nature of the study and the rigorous methods used to measure the sensitivity of ~6000 MET mutants in a pooled format. The dataset generated will be a valuable resource for researchers interested in understanding kinase inhibitor sensitivity and, more broadly, small molecule ligand/protein interactions. The structural analyses are systematic and comprehensive, providing interesting insights into resistance mechanisms. Furthermore, the use of machine learning to define inhibitor-specific fitness landscapes is a valuable addition to the narrative. Although the ESM1b protein language model is only moderately successful in identifying the underlying mechanistic basis of resistance, the authors' attempt to integrate systematic sequence/function datasets with machine learning serves as a foundation for future efforts.

We thank Reviewer 2 for their thoughtful assessment of our manuscript!

Weaknesses:

The main limitation of this study is that the authors' efforts to define general mechanisms between inhibitor classes were only moderately successful due to the challenge of uncoupling inhibitor-specific interaction effects from more general mechanisms related to the mode of ATP-binding site engagement. However, this is a minor limitation that only minimally detracts from the impressive overall scope of the study.

We agree. We have added to the discussion: “A full landscape of mutational effects can help to predict drug response and guide small molecule design to counteract acquired resistance. The ability to define molecular mechanisms towards that goal will likely require more purposefully chosen chemical inhibitors and combinatorial mutational libraries to be maximally informative.”

Reviewer #3 (Public review):

Summary:

In the manuscript 'Mapping kinase domain resistance mechanisms for the MET receptor tyrosine kinase via deep mutational scanning' by Estevam et al, deep mutational scanning is used to assess the impact of ~5,764 mutants in the MET kinase domain on the binding of 11 inhibitors. Analyses were divided by individual inhibitor and kinase inhibitor subtypes (I, II, I 1/2, and III). While a number of mutants were consistent with previous clinical reports, novel potential resistance mutants were also described. This study has implications for the development of combination therapies, namely which

combination of inhibitors to avoid based on overlapping resistance mutant profiles. While one suggested pair of inhibitors with the least overlapping resistance mutation profiles was suggested, this manuscript presents a proof of concept toward a more systematic approach for improved selection of combination therapeutics. Furthermore, in a final part of this manuscript the data was used to train a machine learning model, the ESM-1b protein language model augmented with an XG Boost Regressor framework, and found that they could improve predictions of resistance mutations above the initial ESM-1b model.

Strengths:

Overall this paper is a tour-de-force of data collection and analysis to establish a more systematic approach for the design of combination therapies, especially in targeting MET and other kinases, a family of proteins significant to therapeutic intervention for a variety of diseases. The presentation of the work is mostly concise and clear with thousands of data points presented neatly and clearly. The discovery of novel resistance mutants for individual MET inhibitors, kinase inhibitor subtypes within the context of MET, and all resistance mutants across inhibitor subtypes for MET has clinical relevance. However, probably the most promising outcome of this paper is the proposal of the inhibitor combination of Crizotinib and Cabozantinib as Type I and Type II inhibitors, respectively, with the least overlapping resistance mutation profiles and therefore potentially the most successful combination therapy for MET. While this specific combination is not necessarily the point, it illustrates a compelling systematic approach for deciding how to proceed in developing combination therapy schedules for kinases. In an insightful final section of this paper, the authors approach using their data to train a machine learning model, perhaps understanding that performing these experiments for every kinase for every inhibitor could be prohibitive to applying this method in practice.

We thank Reviewer 3 for their assessment of our manuscript (we are very happy to have it described as a tour-de-force!)

Weaknesses:

This paper presents a clear set of experiments with a compelling justification. The content of the paper is overall of high quality. Below are mostly regarding clarifications in presentation.

Two places could use more computational experiments and analysis, however. Both are presented as suggestions, but at least a discussion of these topics would improve the overall relevance of this work. In the first case it seems that while the analyses conducted on this dataset were chosen with care to be the most relevant to human health, further analyses of these results and their implications of our understanding of allosteric interactions and their effects on inhibitor binding would be a relevant addition. For example, for any given residue type found to be a resistance mutant are there consistent amino acid mutations to which a large or small effect is found. For example is a mutation from alanine to phenylalanine always deleterious, though one can assume the exact location of a residue matters significantly. Some of this analysis is done in dividing resistance mutants by those that are near the inhibitor binding site and those that aren't, but more of these types of analyses could help the reader understand the large amount of data presented here. A mention at least of the existing literature in this area and the lack or presence of trends would be worthwhile. For example, is there any correlation with a simpler metric like the Grantham score to predict effects of mutations (in a way the ESM-1b model is a better version of this, so this is somewhat implicitly discussed).

Indeed we experimented with including these types of features in the XGBoost scheme (particularly residue volume change and distance) to augment the predictive power of the ESM model - see Figure 8 - figure supplement 1; however, we didn't find them as significant. Therefore, the signal is likely very small and/or incorporated into the baseline ESM model.

Indeed, this discussion relates to the second point this manuscript could improve upon: the machine learning section. The main actionable item here is that this results section seems the least polished and could do a better job describing what was done. In the figure it looks like results for certain inhibitors were held out as test data - was this all mutants for a single inhibitor, or some other scheme? Overall I think the implications of this section could be fleshed out, potentially with more experiments.

Figure 8A and the methods section contain a very detailed explanation of test data. We have thought about it and do not have any easy path to improve the description, which we reproduce here:

“Experimental fitness scores of MET variants in the presence of DMSO and AMG458 were ignored in model training and testing since having just one set of data for a type I $\frac{1}{2}$ inhibitor and DMSO leads to learning by simply memorizing the inhibitor type, without generalizability. The remaining dataset was split into training and test sets to further avoid overfitting (Figure 8A). The following data points were held out for testing - (a) all mutations in the presence of one type I (crizotinib) and one type II (glesatinib analog) inhibitor, (b) 20% of randomly chosen positions (columns) and (c) all mutations in two randomly selected amino acids (rows) (e.g. all mutations to Phe, Ser). After splitting the dataset into train and test sets, the train set was used for XGBoost hyperparameter tuning and cross-validation. For tuning the hyperparameters of each of the XGBoost models, we held out 20% of randomly sampled data points in the training set and used the remaining 80% data for Bayesian hyperparameter optimization of the models with Optuna (Akiba et al., 2019), with an objective to minimize the mean squared error between the fitness predictions on 20% held out split and the corresponding experimental fitness scores. The following hyperparameters were sampled and tuned: type of booster (booster - gbtrees or dart), maximum tree depth (max_depth), number of trees (n_estimators), learning rate (eta), minimum leaf split loss (gamma), subsample ratio of columns when constructing each tree (colsample_bytree), L1 and L2 regularization terms (alpha and beta) and tree growth policy (grow_policy - depthwise or lossguide). After identifying the best combination of hyperparameters for each of the models, we performed 10-fold cross validation (with re-sampling) of the models on the full training set. The training set consists of data points corresponding to 230 positions and 18 amino acids. We split these into 10 parts such that each part corresponds to data from 23 positions and 2 amino acids. Then, at each of 10 iterations of cross-validation, models were trained on 9 of 10 parts (207 positions and 16 amino acids) and evaluated on the 1 held out part (23 positions and 2 amino acids). Through this protocol we ensure that we evaluate performance of the models with different subsets of positions and amino acids. The average Pearson correlation and mean squared error of the models from these 10 iterations were calculated and the best performing model out of 8192 models was chosen as the one with the highest cross-validation correlation. The final XGBoost models were obtained by training on the full training set and also used to obtain the fitness score predictions for the validation and test sets. These predictions were used to calculate the inhibitor-wise correlations shown in Figure 8B.”

As mentioned in the 'Strengths' section, one of the appealing aspects of this paper is indeed its potential wide applicability across kinases -- could you use this ML model to predict resistance mutants for an entirely different kinase? This doesn't seem far-fetched, and would be an extremely compelling addition to this paper to prove the value of this approach.

This is exactly where we want to go next! But as we see here, it is going to be hard and require more purposeful selection of chemicals and likely combinatorial mutations to be maximally informative (see also reviewer 2 response where we have added text)

Another area in which this paper could improve its clarity is in the description of caveats of the assay. The exact math used to define resistance mutants and its dependence on the DMSO control is interesting, it is worth discussing where the failure modes of this procedure might be. Could it be that the resistance mutants identified in this assay would differ significantly from those found in patients? That results here are consistent with those seen in the clinic is promising, but discrepancies could remain.

Thank you for pointing this out. The greatest trade-off of probing the intracellular MET kinase (juxtamembrane, kinase domain, c-tail) in the constitutively active TPR system is that while we gain cytoplasmic expression, constitutive oligomerization, and HGF-independent activation, other features like membrane-proximal effects are lost and translatability of some mutations in non-proliferative conditions may also be limited. Nevertheless, Ba/F3 allows IL-3 withdrawal to serve as an effective variant readout of transgenic kinase variant effects due to its undetectable expression of endogenous RTKs and addiction to exogenous interleukin-3 (IL-3).

In our previous study, we were also interested in comparing the phenotypic results to available patient populations in cBioPortal. We observed that our DMS captured known oncogenic MET kinase variants, in addition to a population of gain-of-function variants within clinical residue positions that have not been clinically reported. Interestingly, the population of possible novel gain-of-function mutant codons were more distant in genetic space (2-3 Hamming distance) from wild type than the clinically reported variant codon (1-2 Hamming distance).

For this inhibitor screen, we also carefully compared previously reported and validated resistance mutations across referenced publications to that of our inhibitor screen, and observed large agreement as noted in-text. While discrepancies could definitely remain, there is precedence for consistency.

Furthermore a more in depth discussion of the MetdelEx14 results is warranted. For example, why is the DMSO signature in Figure 1 - supplement 4 so different from that of Figure 1?

In our previous study (Estevam et al., 2024), we more directly compared MET and MET Δ Exon14, and while observed several differences, especially at conserved regulatory motifs, the TPR expression system did not provide a robust differential. Therefore, we hypothesize that a membrane-bound context is likely necessary to obtain a differential that captures juxtamembrane regulatory effects for these two isoforms. For that reason, we did not place heavy emphasis on the differences between MET and MET Δ Exon14 in this study. Nevertheless, we performed parallel analysis of the MET Δ Exon14 inhibitor DMS and provided all source and analyzed data in our GitHub repository (https://github.com/fraser-lab/MET_kinase_Inhibitor_DMS).

In our analysis of resistance, we used Rosace to score and compare DMSO and inhibitor landscapes. We present the full distribution of raw scores in Figure 1 for each condition. However, to visually highlight resistance mutations as a heatmap, we subtracted the scores of each variant in each inhibitor condition from the raw DMSO score, making the heatmaps in Figure 1 - supplement 4 appear more “blue.”

And finally, there is a lot of emphasis put on the unexpected results of this assay for the tivantinib "type III" inhibitor - could this in fact be because the molecule "is highly selective for the inactive or unphosphorylated form of c-Met" according to Eathiraj et al JBC 2011?

The work presented by Eathiraj et al JBC 2011 is a key study we reference and is foundational to tivantinib. While the point brought up about tivantinib's selective preference for an inactive conformation is valid, this is also true for type II kinase inhibitors. In our study, regardless of inhibitor conformational preference, tivantinib was the only one with a nearly identical landscape to DMSO and exhibited selection even in the absence of Ba/F3 MET-addiction (Figure 1E). This result is in closer agreement with MET agnostic behavior reported by Basilico et al., 2013 and Katayama et al., 2013.

While this paper is crisply written with beautiful figures, the complexity of the data warrants a bit more clarity in how the results are visualized. Namely, clearly highlighting mutants that have previously reported and those identified by this study across all figures could help significantly in understanding the more novel findings of the work.

To better compare and contrast novel mutation identified in this study to others, we compiled a list of reported resistance mutations from recent clinical and experimental studies (Pecci et al 2024; Yao et al., 2023; Bahcall et al., 2022; Recondo et al., 2020; Rotow et al., 2020; Fujino et al., 2019), since a direct database with resistance annotations does not exist for MET, to the best of our knowledge. In total, this amounted to 31 annotated resistance mutations across crizotinib, capmatinib, tepotinib, savolitinib, cabozantinib, merestinib, and glesatinib, which we have now tabulated in a new figure (Figure 4) and commentary in the main text:

To assess the agreement between our DMS and previously annotated resistance mutations, we compiled a list of reported resistance mutations from recent clinical and experimental studies (Pecci et al 2024; Yao et al., 2023; Bahcall et al., 2022; Recondo et al., 2020; Rotow et al., 2020; Fujino et al., 2019) (Figure 4A,B). Overall, previously discovered mutations are strongly shifted to a GOF distribution for the drugs where resistance is reported from treatment or experiment; in contrast, the distribution is centered around neutral for those sites for other drugs not reported in the literature (Figure 4C). However, even in cases such as L1195V, we observe GOF DMS scores indicative of resistance to previously reported inhibitors. Given this overall strong concordance with prior literature and clinical results, we can also provide hypotheses to clarify the role of mutations that are observed in combination with others. For example, H1094Y is a reported driver mutation that has been linked to resistance in METΔEx14 for glesatinib with either the secondary L1195V mutation or in isolation (Recondo et al., 2020). However, in our assay H1094Y demonstrated slight sensitivity to glesatinib, suggesting that either resistance is linked to the exon14 deletion isoform, the L1195V mutation, or a cellular factor not modeled well by the BaF3 system.

Finally, the potential impacts and follow-ups of this excellent study could be communicated better - it is recommended that they advertise better this paper as a resource for the community both as a dataset and as a proof of concept. In this realm I would encourage the authors to emphasize the multiple potential uses of this dataset by others to provide answers and insights on a variety of problems.

Please see below

Related to this, the decision to include the MetdelEx14 results, but not discuss them at all is interesting, do the authors expect future analyses to lead to useful insights? Is it surprising that trends are broadly the same to the data discussed?

Our previous paper suggests that Ba/F3 isn't a great model for measuring the differences between MET and MET Δ Ex14, so we haven't emphasized other than to point to our previous paper. We include the full analysis here nonetheless as a resource. Potentially where the greatest differences between resistance mutant behaviors would be observed is in the full-length, membrane-bound MET and MET Δ Ex14 receptor isoforms. While outside of the scope of this study, there is great potential to use the resistance mutations identified in this study as a filtered group to test and map differential inhibitor sensitivities between receptor isoforms.

And finally it could be valuable to have a small addition of introspection from the authors on how this approach could be altered and/or improved in the future to facilitate the general application of this approach for combination therapies for other targets.

See also reviewer 2 response where we have added text.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Major points of revision:

(1) It seems like much of the structural interpretation of the inhibitor binding mode, outside of crizotinib binding, appears to come from docked models of the inhibitor to the MET kinase domain. Given the potential variability of the docked structure to the kinase domain, it would be useful for the authors to consider alternative possible binding modes that their docking pipeline may have suggested. It could also be useful to provide some degree of validation or contextualization of their docking models.

All individual figures are very carefully inspected based on either existing crystal structures of the inhibitor or closely related inhibitors (ATP, 3DKC; crizotinib, 2WGJ; tepotinib, 4R1V; tivantinib, 3RHK; AMG-458, 5T3Q; NVP-BVU972, 3QTI; merestinib, 4EEV; savolitinib, 6SDE). In total, four structural interpretations were the result of docking onto reference experimental structures (capmatinib, cabozantinib, glumetinib, glesatinib). As we wrote above, different conformations and binding modes are possible in predicted mutant structures (as we did here at scale) and included in the ML analysis already.

(2) In the first section, the authors classify an inhibitor as Type Ia on docking models, but mention the conflicting literature describing it as type Ib - it would be helpful to provide a contextualization of why this distinction between Ia and Ib matters, and what difference it might make. It would also be useful to know if their docking score only suggested poses compatible with Ia or if other poses were provided as well. Validation using other method might be beneficial, especially since they acknowledge the conflicting literature for classification. Or at least recontextualization that more evidence would be needed.

Kinase inhibitors have several canonical structural definitions we use to base the classifications in this study. Specifically, type I inhibitors are classified in MET by interactions with Y1230, D1228, K1110 in addition to its conformation in the ATP-binding site. Type I inhibitors are further subdivided into type 1a in MET if it leverages interactions with the solvent front and residue G1163. In prior literature referenced, tepotinib was classified as type 1b, which would imply it does not have solvent front interactions, like savolitinib (PDB 6SDE) or NVP-BVU972 (PDB 3QTI). However, in the tepotinib experimental structure (PDB 4R1V), we observed a greater structural resemblance to other type 1a inhibitors opposed to type 1b (Figure 1 - figure supplement 1b).

(3) The measure used to discuss resistance and sensitivity is ultimately a resistance score derived from the increase or decrease of the presence of a variant during cell growth. This is not a measure of direct binding. It would be helpful if the authors discussed alternative mechanisms through which these variants may impact resistance and/or sensitivity, such as stability, protonation effects, or kinase activity. The score itself may be convolving over all these potential mechanisms to drive GOF and LOF observed behavior.

See the response to the public review. Indeed, our ML framework explicitly included conformational and stability effects as significant in improving predictions.

(4) While it is promising to try and improve the predictive properties of ESM1b, it is not exactly clear why the authors considered their structural data of 11 inhibitors a sufficient dataset with which to augment the model. It would be useful for the authors to provide some additional context for why they wished to augment ESM1b in particular with their dataset, and provide any metrics indicating that their training data of 11 inhibitors provided an adequate statistical sample.

We don't understand what this means. Sorry!

(5) The authors use ESM-1b to predict the fitness impact of each mutation and augment it using protein structural data of drug-target interactions. However, using an XGBoost regressor on a single set of 11 kinase-inhibitor interaction pairs is an incredibly sparse dataset to train upon. It would be useful for the authors to consider the limitations of their model, as well as its extensibility in the context of alternate binding poses, alternate conformations, or changes in protonation states of ligand or inhibitor.

On the contrary - this is 11 chemicals across 3000 mutations. We have discussed alternative interpretations above.

Minor points:

(1) It would also be useful for the authors to provide more context around their choice of regressor. XGBoost is a powerful regressor but can easily overfit high dimensional data when paired with language models such as ESM-1b. This would be particularly useful since some of the features to train on were also generated using existing models such as ThermoMPNN.

Yes - we are quite concerned about overfitting and have tried to assess overfitting by careful design of test and validation sets.

(2) The authors also mention excluding their DMSO and AMG458 scores in the model training and testing due to overfitting issues - it would be useful to have an SI figure pointing to this data.

No - we exclude the DMSO because that is the reference (baseline) and AMG because it has a different binding mode. This isn't related to overfitting.

(3) The authors mention in their docking pipeline that 5 binding modes were used for each ligand docking, but it appears that only one binding mode is considered in the main figures. It would be useful for the authors to provide additional details about what were the other binding modes used for, how different were each binding mode, and how was the "primary" mode selected (and how much better was its score than the others).

The reviewer misinterprets the difference between poses shown in figures, based on mostly crystal structures or carefully selected templates, and the use of docked models in feature engineering for the ML part of the study. Where existing crystal structures do not exist, we performed docking for capmatinib, cabozantinib, glumetinib, glesatinib onto reference structures bound to type I (2WGG) and type II (4EEV) inhibitors. We selected one representative binding mode based on the reference inhibitor, and while not exact, at a minimum these models provide a basis for structural interpretation.

Reviewer #2 (Recommendations for the authors):

My main suggestion is for the authors to add a few sentences (in non-technical language) to the results section, specifically before the results shown in Figure 3, defining gain-of-function, loss-of-function, resistance, and sensitivity. While these definitions are present in the materials and methods section, explicitly discussing them prior to the relevant results would significantly improve the overall readability of the manuscript.

We defined “gain-of-function” and “loss-of-function” mutations as those with fitness scores statistically greater or lower than wild-type. Within the DMSO condition, gain-of-function and loss-of-function labels describe mutational perturbation to protein function, whereas within inhibitor conditions, the labels describe the difference in fitness introduced by an inhibitor.

We have also clarified these definitions where the terms are first introduced: “As expected, the DMSO control population displayed a bimodal distribution with mutations exhibiting wild-type fitness centered around 0, with a wider distribution of mutations that exhibited loss- or gain-of-function effects, as defined by fitness scores with statistically significant lower or greater scores than wild-type, respectively.”

Figure 7D. Please add a bit more detail to the legend on how fold change (y-axis) was calculated.

Here, fold change represents the number of viable cells at each inhibitor concentration relative to the TKI control, measured with the CellTiter-Glo® Luminescent Cell Viability Assay (Promega) as an end point readout. We have updated the legend of Figure 7D with calculation details: “Dose-response for each inhibitor concentration is represented as the fraction of viable cells relative to the TKI free control.”

I must admit, I did not understand what “Specific inhibitor fitness landscapes also aid in identifying mutations with potential drug sensitivity, such as R1086 and C1091 in the MET P-loop” means. These are positions where most mutations lead to greater sensitivity to crizotinib. Is the idea that there are potentially clinically-relevant MET mutations that can be targeted over wild type with crizotinib?

Thank you for highlighting this! The P-loop (phosphate-binding loop) is a glycine-rich structural motif conserved in kinase domains. This motif is located in the N-lobe, where its primary role is to gate ATP entry into the active site and stabilize the phosphate groups of ATP when bound. Therefore, the P-loop is a common target region for ATP-competitive inhibitor design, but also a site where resistance can emerge (Roumiantsev et al., 2002). The idea we’d like to convey is that identifying residues that offer the potential for drug stabilization with the added benefit of having lower risk resistance, is an attractive consideration for novel inhibitor design.

We have added to the text: “Individual inhibitor resistance landscapes also aid in identifying target residues for novel drug design by providing insights into mutability and known resistance cases. This enables the selection of vectors for chemical elaboration with potential

lower risk of resistance development. Sites with mutational profiles such as R1086 and C1091, located in the common drug target P-loop of MET, could be likely candidates for crizotinib.”

Reviewer #3 (Recommendations for the authors):

(1) Suggested Improvements to the Figures:

a) Figure 4A - T1261 seems to be mislabeled

b) In Figure 3A it's suggested to highlight mutants determined to be resistance mutants by this scheme.

c) In Figure 3D it would be informative to highlight which of these resistance mutants have already been previously reported and which are novel to this study

d) Throughout figures 3A, 3D, and 4G the graphical choices on how to highlight synonymous mutations and mutations not performed in the assay needs improvement.

The Green vs Grey 'TRUE' vs 'FALSE' boxes are confusing. Just a green box indicating synonymous mutations would be sufficient. Additionally these green boxes are hard to see, and often edges of this green box are currently missing making it even more difficult to see and interpret.

** In Figure 4A mutants do not seem to be indicated by a line or plus sign, but this is not explained in the legend or the caption. Please add.*

** In 3D and 4G it is not clear if the mutants not performed are indicated at all - perhaps they are indicated in white, making them indistinguishable from scores with 0. Please clarify.*

T1261 and G1242 are now correctly labeled.

In text we have also highlighted reported resistance mutations for crizotinib, which are inclusive of clinical reports and *in vitro* characterization: “These sites, and many of the individual mutations, have been noted in prior reports, such as: D1228N/H/V/Y, Y1230C/H/N/S, G1163R.”

We have adjusted the heatmaps to improve visual clarity. Mutations with score 0 are white, as indicated by the scale bar, and mutations uncaptured by the screen are now in light yellow. The green outline distinguishing WT synonymous mutations have also been adjusted so edges are no longer cut off. In our representations, we only distinguished mutations by the score color scale bar and WT outline. What looked like a “plus” or “line” in the original figure was only the heatmap background, which now should be resolved in the updated figure and legends for Figure 3 and Figure 4.

(2) Some Minor Suggested Improvements to the Text:

a) The abbreviation CBL for 'CBL docking site' is used without being defined.

b) Figure 3G is referenced, but it does not exist.

c) In the sentence 'Beyond these well characterized sites, regions with sensitivity occurred throughout the kinase, primarily in loop-regions which have the greatest mutational tolerance in DMSO, but do not provide a growth advantage in the presence of an inhibitor (Figure 1 - Figure Supplement 1; Figure 1 - Figure Supplement 2)'. It is not clear why these supplemental figures are being referenced.

d) *In the supplement section 'Enrich2 Scoring' has what seem like placeholders for citations in [brackets]*

Cbl is a E3 ubiquitin ligase that plays a role in MET regulation through engagement with exon 14, specifically at Y1003 when phosphorylated. This mode of regulation was more highlighted in our previous study. However, since Cbl was only mentioned briefly in this study, we have removed reference to it to simplify the text.

In addition, we have removed the figure 3G reference and corrected the in-text range. We have also removed references to figure supplements where unnecessary and edited the “Enrich2 scoring” method section to now reference missing citations.

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